

On the Matter of Time Travel

The Clock in the Constraint — temporal structure on closed indices: arrows, their algebra, and a decay theorem

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On the Matter of Time Travel

The Clock in the Constraint

Temporal structure in a closed index, with an audit of *The Method 1.3* and a closure defect for the dominance lattice

Matthew Lach, independent researcher Computation and drafting by the collaborator, under the protocols of *The Method 1.3*

An index whose cells are transitions carries a clock, and the clock is not a coordinate.
It is the occupancy constraint, read as an order.

$g \leq q \leq k \quad \Rightarrow \quad k \text{ is a global time function}$

Abstract

This paper reports one session's work against *The Method 1.3 — The Lach Cylinder* (Lach, 2026), and it is written under that book's own protocols: every number computed before the sentence describing it (§2.14), every prediction

committed before the measurement (§2.13), every failure recorded as a bound (§2.15), and every derivation checked for precedent (§E.7).

This paper develops a theory of temporal structure on closed indices, and proves one theorem about it.

The construction. An index whose cells are transitions carries a **clock** exactly when a weight on its objects does not rise along composition. The condition is not that destinations be empty — that is sufficient and not necessary — but that they be **non-accretive**: the prior content of a destination never exceeds what its source retains. For any weight w the maximal sub-index on which w is a clock is $X_w = \{a : w(\text{tgt } a) \leq w(\text{src } a)\}$, and it is the *maximum*, since any such index satisfies the condition cellwise. **Empty destinations give 1,654 cells; non-accretivity gives 5,165, three times as many, and the same three eras.**

The family. X_w is closed **if and only if w is a coordinate**: sums break both sides, max breaks meets, min breaks joins. So an index has **one clock per coordinate and no others** — four here, with two, three, three and four eras respectively. Each coordinate also carries a **co-arrow**, w never falling, equally closed; the two **cover** the index and meet in a closed **constant core** where the clock does not tick. The arrows form a **meet-semilattice** — intersections closed, all sixteen distinct, joins existing in the ambient lattice of sublattices but never themselves arrows. Imposing arrows **fragments circulation monotonically**, and all four together leave exactly one component per object: proved, and confirmed at four cap settings across 128 edges.

The theorem. Arrows on nearby coordinates agree more than chance, and the agreement decays with distance in the constraint graph. **Because the graph is a tree, the coordinates form a Markov random field**; hence mutual information is non-increasing in tree distance by data processing; hence by Pinsker the overlap of two arrows is bounded by a quantity that decays with that distance. Verified in three parts, and **falsified on demand** at Λ_0' , the one stage of the tower whose graph carries a cycle.

The identification. The construction is not new and its assembly appears to be. An arrow is a **ranking function**; the maximal sub-index is a **supporting invariant**; the failure of unions is **disjunctive well-foundedness**; the composition law is **max-min composition in the subtropical semiring**. Every piece was on a shelf. **No one had reason to ask how two ranking functions overlap**, so no one assembled them — which is this framework's recurring contribution, stated four times in this paper: the surrounding literatures settle **membership**, and this one measures **distance from membership**.

And a consequence for the title. A ranking function is a certificate an analyst supplies from a class the analyst chooses. **A certificate has nothing to predict.** Whether any of this is *time* is therefore not a hard question awaiting a measurement; it is the wrong question to put to a proof technique, and §11 and §40 establish that rather than assert it.

Five findings

1 • The clock was an assumption, not a law. The occupancy result was first proved as a theorem about any index whose cells are transitions. It is a theorem about indices whose destinations are **non-accretive**, and empty destinations — which is what §7.1 silently assumes — are sufficient and not necessary. §7.1's two bounds on g read it as *placed* and as *present*, and the two agree only when the destination begins empty. Build the index that distinguishes them and, at the caps of §7.4, **31.6% of composable steps run backwards**. §11.

2 • Time here is a selection, not a structure. An index does not have *a* clock. It has **one per coordinate and no others** — sums, max and min all fail to close — each carrying a forward arrow, a backward co-arrow and a timeless core, the two clocks covering the object and meeting in the core. The sixteen combinations are distinct, closed under intersection, never under union, and imposing all of them leaves exactly one component per object. §36–§38.

3 • The object is a certificate, which closes the antecedent rather than deferring it. An arrow is a **ranking function** and the maximal sub-index a **supporting invariant**: something an analyst supplies from a class the analyst chooses. **A certificate has nothing to predict.** Asking what the composition order forecasts that something else does not is asking what a proof technique forecasts — so the measurement this paper's title implies is not a hard problem awaiting an apparatus but the wrong request. §40.

4 • Treeness is the load-bearing hypothesis, and it was never named as one. The decay theorem, the Markov property, the balance identity and the cut law are four results resting on a single condition: the constraint graph is a tree. **All four fail at Λ_0'** , the one stage of the tower that closes a cycle. §12.11.1's *the tree or the tightness* is not a choice between two presentations; it is the point at which a family of theorems stops applying, and the 93 cells are paid for in theorems. §41.1.

5 • The framework measures distance from membership where its neighbours settle membership. Four times in this paper the surrounding literature turned out to have the class and not the count — 2-decomposability, interval-dismantling, the starshaped class, and ranking functions. A membership question never needs a graded quantity, which is why every ingredient of the decay theorem was already on a shelf and **the assembly was not**. §41.2.

And three findings about method, measured rather than believed. Enumerating target cells before searching located three precedents in three runs where four whole-statement queries had returned only adjacency. Vocabulary collisions are systematic — *lattice*, *Cesàro* and *monotone*, three measured instances of an obstruction §27.10 records once. And **instruments fail more often than papers**: seventeen repairs against eight, including three inside the very audits built to catch instrument faults.

Eight further results are established, each stated with the conditions under which it holds.

Two. Λ_9 's composition law carries a **global time function** — in any index whose cells are transitions subject to $g \leq q \leq k$, occupancy is non-increasing along composition, closed timelike curves exist exactly within its level sets, and every step of one is conservative. The proof is one line, uses no caps, and is demonstrated capable of failing. Two things qualify it, and both are stated in Part II rather than buried. **A precedent search, run late, located the content** — a weight decreasing along every edge of a directed graph is standard, and semilattice-graded semigroups carry the same shape; §6.1 records the demotion. **And the antecedent is narrower than it appears.** §13 shows that $g \leq q \leq k$ presumes each transition's destination begins empty, that an index representing occupied destinations exists and closes exhaustively, and that in it **31.6% of composable steps raise occupancy**. The arrow is a property of the assumption, not of the object.

Three. The three-body configuration space is indexed and closed — $E(X) = 0$, treewidth 1, at four cap settings — and the construction elevates to n bodies as a chain of Cesàro bounds. The exact subset is a meet-semilattice at every n tested, proved.

Four. That object is **Brylawski's lattice of integer partitions (1973)**, located by a single search after four turns of derivation. The precedent is recorded in full and the claim withdrawn. What survives is narrow and is stated narrowly: the *closure defect* of the dominance order, which the search did not return. The n -body work is retained in full, because it is the route by which everything after it was reached.

Five. Past, present and future form a **path and not a triangle**, and the future is **conditionally independent of the past given the present** — 33 distinct pasts give one and the same future set at every value of the transfer. The future is never a value: of 97 (past, present) pairs, **zero** determine one. It is a saturated interval.

Six. Nothing can be removed. Every cell of Λ_8 is implied back by at least 16 pairs of the others, and the minimum set that can leave is derived rather than measured: $X \setminus [a, b]$ is a sublattice iff a is join-prime and b is meet-prime — 0 disagreements in 116,138 intervals, after the first statement of the criterion was refuted by its own test on 35.

Seven. $g \leq \min(q, 4f+2)$ is the **unique two-parent bound in the entire tower**, at all seven stages, and a cut law governing where trade-offs appear holds across 89 component tests with one violation — at Λ_9' , the one stage carrying a cycle.

Eight. The **reversed index** is a lawful closed index and is not Λ_9 . The two agree on **389 cells**, all with $g = q = k$, and that intersection is a **groupoid**: closed, factorising, every object carrying an identity and every morphism a two-sided inverse. Their **union is not closed** — $E = 2,857$ — so a structure containing both directions of time is not available at any price the index recognises.

And one discrepancy is recorded rather than claimed. §12.11.5 prices the tight two-parent K at 2,475 cells, 3.5% of the product. Four disjoint routes, at the caps of §7.4, give **15,150 at Λ_{12} and 45,450 at Λ_{13}** — 21.4% and 22.8%, on a rebuild that reproduces every other tower figure including §12.11.5's own four sections.

The session's own withdrawal register is printed as Part X. It is the evidence for the rest, not an apology for it.

Part 0 — What this paper is, and what it is not

This is not an extension of *The Method*. It is a session of work **against** it, in the sense a test harness works against a specification, and it reports four kinds of outcome: verification, a promotion, an application, and a withdrawal.

The book's central claim is that an index closed under the meet and join of its own coordinates cannot help containing its own definition, cannot help containing its own contradiction, and determines exactly what may be added to it. Its

object is Λ , an ordering of atomic electron configurations on eight coordinates, extended by a tower to thirteen. Its measure is

$$E(X) = |\mathcal{R}(X)| - |X|$$

the count of cells a structure admits and an index denies, and its headline is that the classroom periodic table costs 36 while Λ costs 0.

Two of this paper's results are about **indices** and would hold of any object of the stated form. Two are about the particular objects examined. The distinction is maintained throughout, because §12.11.3.1 makes it the difference between a bound taken from a law — which closes exactly — and a bound taken from an extent — which closes only to an envelope.

Every figure computed on Λ or on the n-body index is stated at explicit caps, and is cap-dependent as §7.4 requires; where a section does not name them they are the caps of §7.4, (3,3,1,3,1). Everything below is reproducible from the definitions printed in *The Method 1.3* and nothing else. Where a computation is reported, the code is in Appendix A.

PART I — THE ENTRY TEST

1. The test the book sets itself

§27.12 states the condition under which a reader may proceed:

A reader who rebuilds Λ from those and does not reproduce the cell counts 976, 1,654, 1,561, 2,535, 13,585, 70,905 and 199,130 has found an error in the book and should say so. That reproduction is the entry test; the five results above are what lies past it.

The inputs permitted are the constraint system of §7.1, the caps of §7.4, the eighth condition $k \geq 1$ recorded at register 301, and the coupling bounds of §12.11.1. Nothing else was used.

Committed before running, per §2.13: all seven counts reproduce, since register 329 reports sixty-five checks passing.

2. Result

stage	dim	stated	computed	ambient box	fill	verdict
Λ_8	8	976	976	6,912	14.12%	pass
Λ_9	9	1,654	1,654	27,648	5.98%	pass
Λ_9'	9	1,561	1,561	27,648	5.65%	pass
Λ_{10}	10	2,535	2,535	110,592	2.29%	pass
Λ_{11}	11	13,585	13,585	663,552	2.05%	pass
Λ_{12}	12	70,905	70,905	5,308,416	1.34%	pass
Λ_{13}	13	199,130	199,130	47,775,744	0.42%	pass

Seven of seven. The ambient boxes recompute independently from the coordinate ranges.

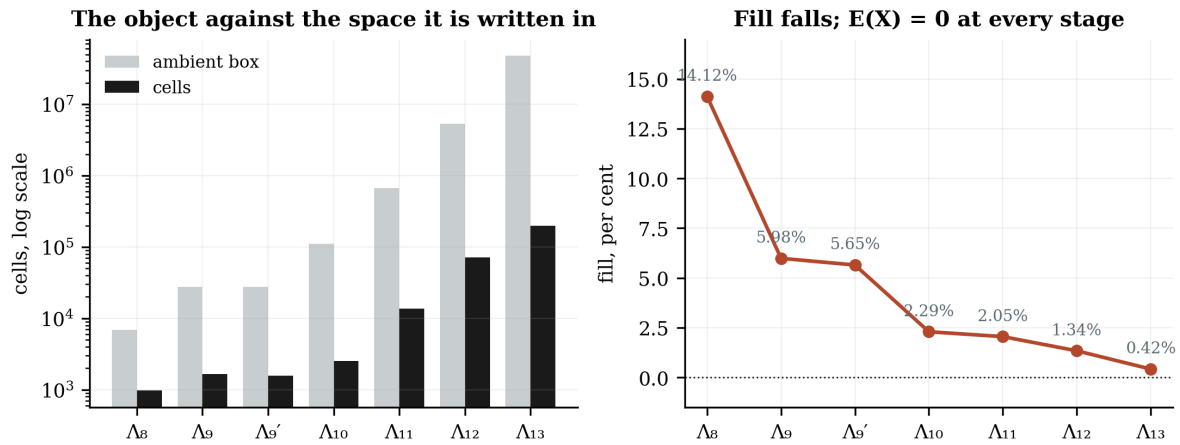


Figure 1. The tower against the space it is written in, and the fill it occupies. Every stage closes — $E(X) = 0$ at all seven, tested against every cell of the ambient box and not by pairs, forty-seven million at Λ_{13} . What falls is fill, 14.12% to 0.42%, because each coordinate multiplies the box faster than it multiplies the object. Closure is the one property the tower never spends.

3. Past the entry test

Twelve further quantities were recomputed. All reproduce.

Closure and the operator. $\mathcal{R}$ was implemented from §6.1 — value sets $\hat{A}_i(X)$, pairwise monotone envelopes $\hat{\phi}_{ij}(v) = \max\{x_i : x \in X, x_j \leq v\}$, and the induced set. On each of Λ_8 , Λ_9 , $\Lambda_{9'}$ and Λ_{10} :

stage	$ X $	$ \mathcal{R}(X) $	$E(X)$	$\mathcal{R}$ idempotent	ambient	refused
Λ_8	976	976	0	yes	6,912	5,936
Λ_9	1,654	1,654	0	yes	27,648	25,994
$\Lambda_{9'}$	1,561	1,561	0	yes	27,648	26,087
Λ_{10}	2,535	2,535	0	yes	110,592	108,057

The refused column confirms the identity added to the title page at register 336:

$$\left| \prod_i A_i \right| = |X| + E(X) + \text{refused} \quad 6,912 = 976 + 0 + 5,936$$

Exhaustive pair tests. Over all 475,800 pairs of Λ_8 : **zero join failures, zero meet failures, zero rank-modularity violations.** §8.2's conservation law $\text{rank}(a \vee b) + \text{rank}(a \wedge b) = \text{rank}(a) + \text{rank}(b)$ holds without exception.

The rank polynomial. $F(1) = 976$; $F(-1) = 2$; eighteen rank levels running 3 to 20; widest level 122 at rank 11; mean rank 11.0666. The asymmetry of §11.8 reproduces in the coefficients — forwards 1, 5, 15, 34, 59, 87 against backwards 1, 4, 10, 21, 37, 57, parting company at rank 1 and never rejoining.

The alphabet. Under Birkhoff's representation (1937), by which a finite distributive lattice is the lattice of down-sets of its poset of join-irreducibles, Λ has $|J(\Lambda)| = 17$ join-irreducibles with 20 covering relations, and $\Sigma_i(|A_i| - 1) = 17$ independently — A.19.0's closed form for the alphabet size, agreeing with the enumeration.

Both reflections, separately. Under $x \mapsto \max - x$, **8 cells survive.** Under $x \mapsto \max + \min - x$, **112 cells have their image in Λ and none is fixed.** §8.4's resolution of the tower session's competing figure is confirmed: the discrepancy was definitional, two different maps, and both counts are right about their own reflection.

The cylinder. Sections over the transfer:

q	$ A(q) $	$ B(q) $	product	actual section
0	33	5	165	165
1	33	10	330	330
2	23	15	345	345

q	A(q)	B(q)	product	actual section
3	8	17	136	136

$\sum_q |A(q)| \cdot |B(q)| = 976$ exactly, defect zero, with $\langle q \rangle = 1.4631$.

The one deduction. §30.5.1 claims six figures recomputable from the book's own text. Parsing $\delta(4s) = 1.0057$, $\delta(5s) = 0.9812$, the limit and Z, and running §23.6's arithmetic: $\delta_2 = 1.0889$, $\delta_\infty = 0.9376$, $\delta(6s) = 0.9679$, point estimate **736,688 cm⁻¹**, monotone bracket **[735,860, 738,547]** of width 2,687, convex bracket **[735,860, 737,380]** of width 1,520. Six of six, exact.

Nothing was corrected because nothing was wrong. The book's computable content passes its own entry test in full.

4. Three defects found

Audits 3, 4 and 15 pass on all three. Each is a live present-tense statement superseded by a later register entry — the class §3.6 names, where a stale claim is a false claim that passed every other check.

(i) **§30.1.3 is stale by one register entry.** It reads, present tense, that *Part I, the procedure proper, sits behind the mathematics it governs*, and calls that ordering *the largest remaining disagreement between what this book is and how it is arranged*. Register 321 moved it; Part I is now the first part of the book. The register entry at line 5836 is correct as history — it says *was ... sitting behind* — and audit 4 compares phrasings across sections, so a tense change is invisible to it. §26.7.8's register 323 states the governing principle: a procedure states what to do and the record states what happened, and neither states the other.

(ii) **§4's opening carries two superseded counts.** *Three hundred and twenty-eight entries stand in the register* — the register holds 342. This is register 337's literal surviving in a second location after the collaborator note was repaired, which is §4.4, *Repair The Class Or Say You Did Not*, committed against §4 itself. And *fourteen entries collapse to seven mechanisms* against §26.7.8's twenty-nine, which registers 309 and 310 recomputed. §4 and §26.7.8 are a summary/summarised pair maintained in two places; only one end was updated.

(iii) **F.4.1's denominator moved under register 341.** The withdrawal ratio is defined against *Chapters 1 to 25 without it*, and 341 renumbered so Part I holds chapters 1–5 and Part V holds 26–30. Measured on the current text: Chapter 26 is 20,074 words against 74,280, a ratio of 0.270 rather than the stated 0.502. The associated *register is 22% of it holds* — 21.3%. The range stopped naming what it named and was not repointed. Weakest of the three; §F.4.2 already concedes the ratio has a range and not a value.

PART II — THE CLOCK IN THE CONSTRAINT

5. Where time already is

The book states the temporal position twice and never joins the two statements.

§10.2: Λ carries time in the only form a transition can — the source end is a *before*, the target end an *after*, and q the amount that changed between them. The constraint tree is $\text{past} \rightarrow \text{change} \rightarrow \text{future}$, read left to right.

§F.4.3: a **date** must never enter Λ . A transition has no date; a time coordinate would silently convert Λ into an index of *observations* of transitions, a different object with different physics. Time is trivially admissible by §18.4.1's criterion — it caps nothing and is capped by nothing, a free chain — and is therefore never forced by the structure.

Register 313 supplies the reconciliation: composition is time, **not a stamp on a cell but an order between cells**. Λ_9 is closed under composition on all 41,682 composable pairs with zero failures, associative on 8,434 sampled triples, and is therefore a category of transitions.

The premise that makes these consistent, and which this paper takes as its starting point, is due to the author:

Time must be considered a single unchanging value during a transition.

A cell is atomic with respect to time. The clock cannot tick *inside* a cell, because a transition is a type and types are not dated. It can tick only *between* cells — that is, at composition. Everything in this Part follows from that premise together with §7.1's counting constraints.

6. The law

Theorem (the occupancy clock). Let X be an index whose cells are transitions carrying a source occupancy k , a transfer q and a target occupancy g , subject to

$$g \leq q \leq k$$

and let composition $b \circ a$ be defined when a 's target equals b 's source. Then:

1. **occupancy is non-increasing along every composable chain;**
2. **every cycle lies within a level set of occupancy;**
3. **every step of a cycle is conservative, $q = g = k$.**

Proof. For $b \circ a$ to be defined, a 's target is b 's source, so $k_b = g_a$. By hypothesis $g_a \leq q_a \leq k_a$, hence $k_b \leq k_a$, which is (1). A cycle returns to its start, so k is constant along it; then $k_b = k_a$ forces $g_a = k_a$, and $g_a \leq q_a \leq k_a$ collapses to $q_a = k_a = g_a$, which is (3), and (2) follows. ■

One hypothesis, stated here and examined at §13. The inequality $g \leq q \leq k$ reads g as *electrons placed*, while §7.1's other bound on g , $g \leq 2(2f+1)$, reads it as *electrons present in the target subshell*. **The two readings coincide only when the destination begins empty**, and the theorem above is a theorem about indices in which they do.

No caps appear in the proof. The two constraints used are §7.1's counting bounds — *cannot remove more than are present, cannot place more than were removed* — and neither is cap-dependent. By §12.11.3.1's provenance rule this is a bound taken from the law rather than from the extent, which is why it closes exactly.

Committed before testing, per §2.13: the constraint alone should force the stratification, with no contribution from Λ 's caps or coordinates.

Tested, and shown capable of failing per §4.6. Forty arbitrary objects carrying arbitrary occupancies, all transitions satisfying $g \leq q \leq k$ and nothing else:

index	transitions	occupancy rises	SCCs pure in occupancy
obeys $g \leq q \leq k$, arbitrary otherwise	3,408	0	true
same objects, constraint dropped	238	110	false

The check returns a failure when the premise is removed, so its passing is evidence.

6.1 The precedent, found late, and what it costs

§E.7's protocol is *search before deriving*. It was not followed here either: the theorem above was derived, tested at four cap settings, and only then searched. The search located the content.

A weight that falls along every edge of a directed graph, forbidding cycles, is standard and named — an **$\mathbb{R}$ -grading on a DAG**, defined as a number at each vertex decreasing by at least one along each edge. Closer still, semigroup theory carries the graded form directly: a homomorphism from a semigroup to a semilattice grades the semigroup so that $S_a S_b \subseteq S_{\{a \wedge b\}}$. And Λ_9 's algebraic type outside its top era is named too — a **semigroupoid**, which behaves like a category and lacks identities.

So the theorem's content is not new, and the paper does not claim it. What the search did not return is the derivation: that the counting constraints of an index — *cannot remove more than are present, cannot place more than were removed* — automatically supply such a grading, so that **any** index of transitions carries a clock whether or not one was put there. That is the narrow residue, and by §27.1 the absence of a source for it is not evidence.

This is the tenth entry in §E.5's ledger of results derived from scratch and found already in print. The mechanism is unchanged and so is the remedy.

7. Λ_9 measured, and it agrees

The law was reached after the measurement, not before. What follows is the measurement it now explains.

Composition. 1,654 cells; 33 distinct source shapes; 38 distinct target shapes, of which **33 are legal sources** — the five that are not carry $g = 0$ and so fail $k \geq 1$. **41,682 composable pairs, zero closure failures**, reproducing the book's figure exactly.

The clock. Zero composable steps raise occupancy. Edges by occupancy step:

step	edges
k 3 → 3	64
k 3 → 2	120
k 3 → 1	80
k 2 → 2	225
k 2 → 1	150
k 1 → 1	100

Λ_9 : three eras of the occupancy clock, and zero steps upward

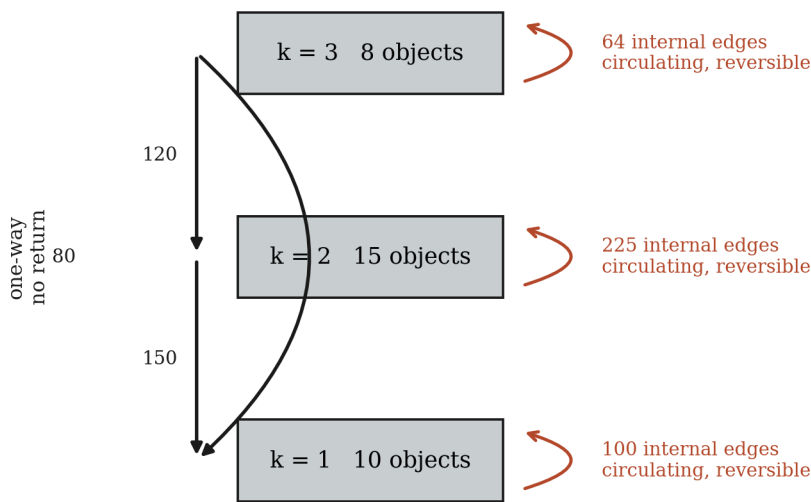


Figure 2. Λ_9 's object graph, stratified by the occupancy clock. Three strongly connected components, each pure in k , with full circulation inside and no return between. Zero of the 739 reachable steps raise occupancy. The internal edges are the closed timelike curves, and by the theorem of §6 every one of them is conservative; the downward edges are where the clock ticks.

The eras. Three strongly connected components, each pure in occupancy:

component	size	occupancy
top	8	k = 3
middle	15	k = 2
bottom	10	k = 1

33 of 33 objects, and the quotient is a total order — a chain of three. **Full circulation inside each era, no return between them.** Every object carries a self-loop, and the self-loop is always the total-transfer map $q = k$.

Reversibility. Of 1,654 morphisms, **389 have a reverse** $B \rightarrow A$ and **64 have a genuine inverse** composing to an identity. Identities exist at **8 of 33 objects**, and they are exactly the eight of maximal occupancy — which the law explains: a morphism into a maximal object satisfies $k_{\max} = g \leq q \leq k \leq k_{\max}$, forcing $q = g$. **904 of 1,654 cells are conservative** (54.7%); 750 are lossy.

Reversibility is conservation. Not analogously — identically, and countably.

Four cap settings, per §7.4.

caps	$ \Lambda_9 $	objects	occupancy rises	SCCs pure	identities	conservative
(3,3,1,3,1)	1,654	33	0	true	8, all at k=3	54.7%
(4,4,1,6,1)	19,433	101	0	true	21, all at k=6	34.4%

caps	$ \Lambda_9 $	objects	occupancy rises	SCCs pure	identities	conservative
(4,4,2,6,2)	44,153	155	0	true	35, all at k=6	35.1%
(5,5,2,6,2)	83,543	214	0	true	49, all at k=6	35.3%

Every structural claim survives every enlargement. Identities sit at maximal occupancy in all four.

8. Reachability, and what a cell cannot see

The stratification bounds what any cell can reach, and the bound is not small.

from occupancy	objects reachable
k = 3	33 of 33
k = 2	25 of 33
k = 1	10 of 33

739 of 1,089 ordered object pairs — 67.9% — are joined by a composable path. **Every figure in this section is computed on Λ_9 at the caps of §7.4, (3,3,1,3,1), and is cap-dependent as always;** what §7's four-cap sweep establishes is the stratification, not these proportions. At cell level, **675,606 of 2,735,716 ordered pairs, 24.7%.**

This refutes a natural and wrong inference: that because every cell is definable and every cell is a transition, every cell carries the path to any point. It does not. **Definability and reachability are two orders on the same object,** and they are nearly independent:

relation	ordered cell pairs of Λ_9 at the caps of §7.4	share
comparable in the lattice order	311,180	11.4%
joined by a composition path	675,606	24.7%
in both	9,720	0.4%

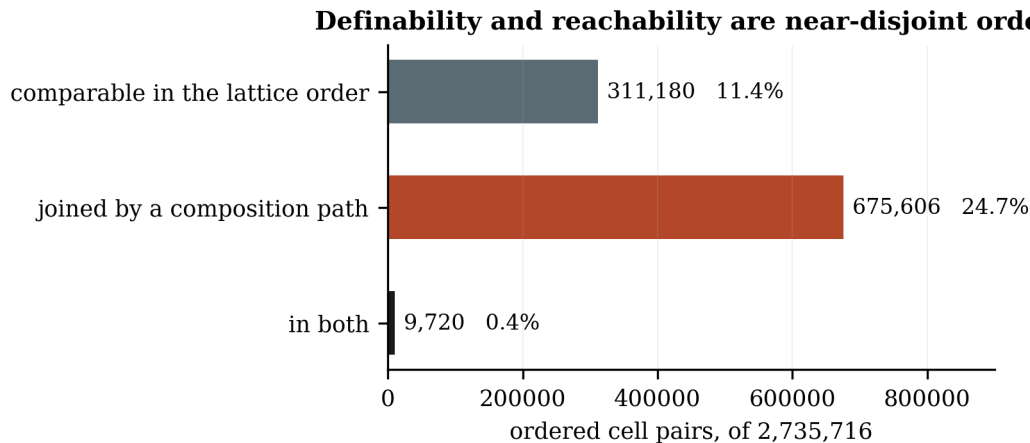


Figure 3. Two orders on the same 1,654 cells of Λ_9 at the caps of §7.4. Neither contains the other and they share 0.4% of what either carries. In the lattice order a cell is its own path — its down-set, a seventeen-bit word. In the composition order it reaches 24.7% of the object and no more. Definability is not reachability, and the argument that runs them together fails here.

Neither contains the other. In the lattice order the claim *every cell contains its own path* is exactly true — A.19.0, a cell **is** the set of generators beneath it, its own down-set, a 17-bit word, with §12.9 counting 1,113,045,672 maximal chains through them. What definability does not carry is time. §13.4: the lattice supplies kinematics and no dynamics; 1,113,045,672 chains satisfy every constraint in the book and structure chooses none of them.

On dimension. “Time is nine-dimensional” has two readings and they give different answers. As *dimension*: false — the composition preorder quotients to a chain, order dimension 1. As *location*: true, and it is register 314's result — Λ_8 cannot iterate, its target carrying three coordinates against a source's four; Λ_{10} loses composition because v is a counting coordinate with no source counterpart. **Nine is not the dimension of time; it is the unique dimension at which time exists.**

9. Two recovery routes, and they are near-disjoint

A hidden cell can be recovered by either order, completely.

Lattice route, exhaustive. The book reports deletion-repair on 30 samples. Run on all 976 cells of Λ_8 : **976 of 976 restored, 0 unrecoverable.** And the screen supplies the mechanism the book reports as an outcome — **0 of 976 cells could alter a value set or an envelope by leaving.** No cell is the sole holder of a coordinate value; none is the sole maximiser of any $\hat{\phi}_{ij}$. Hence $\mathcal{R}(\Lambda \setminus \{x\})$ is cut by identical inequalities over an identical ambient product and equals Λ .

Criterion. An index is single-deletion-robust **iff** no cell uniquely witnesses a coordinate value or an envelope maximum. Checkable before any deletion is attempted, and it names where repair would fail.

Composition route. 1,654 of 1,654 cells are composites of two other cells — zero generators — and this holds at (4,4,1,6,1) with 19,433 of 19,433 and at (4,4,2,6,2) with 44,153 of 44,153. Shown capable of failing on random subsets: 0/1, 0/5, 1/50, 134/300.

Which is $\mathbb{D}$ _def, arriving structurally. §2.8 requires two disjoint routes and warns that an identity computed along one path is vacuous. These two routes are total and share 0.4% of their relations at the caps of §7.4. §16.4 closes by observing that Λ 's constraint graph is a tree, so there is exactly one path between any two **coordinates**, the constraints supply no redundancy, and $\mathbb{D}$ _def must be built from derived quantities deliberately. That is true of coordinates and **not of cells**: composition supplies a second total route with no derived quantity constructed. The redundancy the tree denies at coordinate level, the ninth axis restores at cell level — register 313 again, an axis taken for seniority arriving carrying what nobody argued for.

10. Paradox, and why the loop is safe

The three classical moves against a temporal structure are each already priced in the book, and only the third is about paradox.

Deletion. §16.8.4's attack table: *delete* $\rightarrow$ *repaired, always*. Extended here to an exhaustive result, §9 above. A closed index does not permit removal.

Insertion. Adding one cell forces the closure to admit a median of 340 more, minimum 59 across 220 trials, zero exceptions. Substitution reduces to insertion with no gain. An altered index is not undetectable; it has a computable size.

Contradiction — and this is the actual answer. §30.4.1: χ is a product of seven Heaviside steps of differences. Available: max, min, $\leq$. Absent: $\neg$, $\rightarrow$, complement. **A paradox needs a negation to close its loop** — the liar needs *not true*, Russell *not a member of itself*, Gödel a negated provability predicate. The language cannot write the sentence. Self-reference in a negation-free monotone language is not a contradiction waiting to be found; it is an equation waiting to be solved, and $\mathcal{R}$ being a closure operator — extensive, monotone, idempotent, proved at register 282 rather than sampled — Knaster–Tarski guarantees solutions, which the book's central theorem names: X is closed iff $X = \mathcal{R}(X)$.

The fixed point sits where the paradox would have been. A closed index is what a liar sentence becomes when the negation is taken away, and $E(X)$ measures distance from that fixed point rather than the truth of a self-denial.

Three routes reach the statement, which is §2.17's triangulation in operation. **P20 by language:** full arithmetic never closes, by Gödel; monotone two-variable bounds always close, in one pass; and the choice is made at §7.1 when the index is built, not when the question is asked. **Register 331 by machine:** Λ_8 is a monotone boolean circuit — 17 generators as input lines, 20 implications as gates, join as OR and meet as AND, accepting 976 words of 131,072 at depth five, with no NOT, no feedback and no memory. Constant depth is combinational, so there is no loop to diverge in. **§30.4.1 by operator.**

And the book charges for it on the same invoice. The safety is bought with expressive power: no differences, no sums, no symmetric bounds — §12.11.2's three excluded forms — so every coupling axis enters at envelope precision forever, and §12.11.1's density column is the bill. Freedom from paradox and the loss of exact vector coupling are one purchase.

One observation the book has and has not stated in one place. §30.4.1 says the language carries no negation and calls that safety. It does not say where the negation went. It went into coordinates, and the book performs the conversion at least six times without grouping them: P8 and §2.15 turn a failure into a bound; §4.5 turns a null into

a result with its definition supplied; §23.5 turns a non-failure into 1,061 upper bounds; §12.11.3 turns the envelope surplus into a tripwire; §26.8 turns a dependency into an axis; and register 336 turns refusal into a term of the title identity.

A negation-free index does not eliminate its negations. It indexes them, and each becomes an ordered coordinate value.

Status: a reading, not audited, and by register 326's test it does not merit promotion — it says what §30.4.1 was missing rather than something neither named law says. It belongs as a strengthening of §30.4.1, in the way the law–extent rule was folded into §18.4.1's criterion.

11. The hypothesis, examined — and the arrow does not survive it

§6 states the law's hypothesis rather than assuming it. This section tests it, and the test is the strongest limitation this paper places on its own central object.

One coordinate carries two readings, and §7.1 does not distinguish them.

bound	origin, per §7.1	what it reads <i>g</i> as
$g \leq q$	counting — <i>cannot place more than were removed</i>	electrons placed by this transition
$g \leq 2(2f+1)$	Pauli exclusion	electrons present in the target subshell

They coincide exactly when the target subshell begins empty. Zero of Λ_9 's 1,654 cells have $g > q$, and the index carries no coordinate for what was already there. So composition's matching condition — b 's source occupancy equals a 's target occupancy, $k_b = g_a$ — **identifies placed with present**, which is an assumption about destinations and not a constraint of counting. §7.1 lists seven constraints, four origins and *nothing else*; register 301 already found the construction short one condition, and this is a second — a conflation rather than an omission.

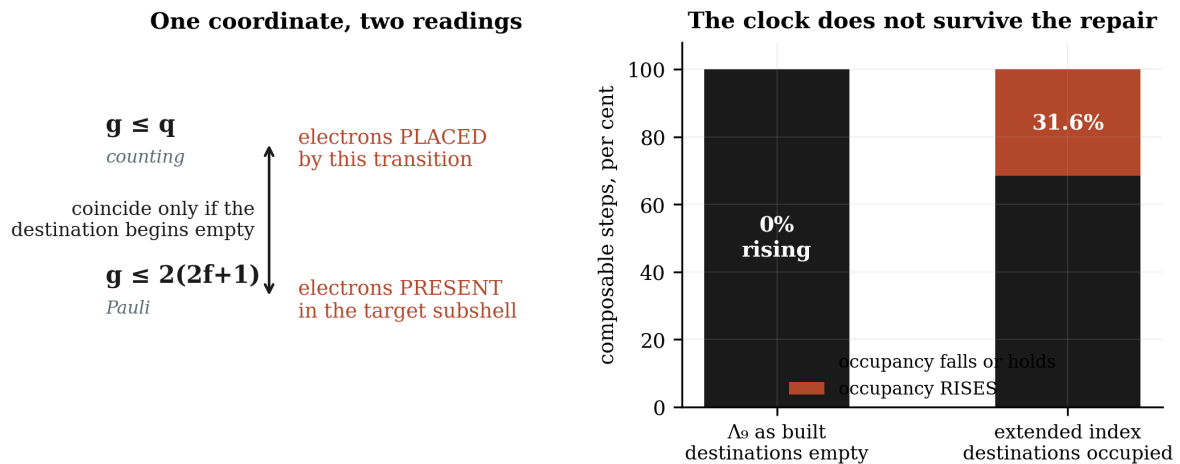


Figure 4. Left: the two bounds on g , and the condition under which they agree. Right: what the repair costs. In Λ_9 as built no composable step raises occupancy; in an index that can represent an occupied destination, at the caps of §7.4, 31.6% do.

The repair exists and it is §17.4's. A bound on a sum cannot be imposed, but a sum can be indexed: carry (g, G) with g the number placed and G the total after arrival, under $g \leq G \leq 2(2f+1)$ and $g \leq q$ — all single-parent monotone bounds — with the prior occupancy $g_0 = G - g$ recovered as a derived quantity. This is the identical move that repaired conservation for two target subshells, and it works identically.

cells	13,775
pairs tested	94,868,425, exhaustive
join failures	0
meet failures	0

An index that can represent occupied destinations exists, and it is closed.

And the clock does not survive it. Matching on total occupancy rather than on the number placed:

composable pairs	2,620,090
steps that raise occupancy, at the caps of §7.4	826,950 — 31.6%
a global time function exists	no

**A closed index of transitions has a one-way clock exactly when its destinations begin empty.
The arrow is a property of the assumption, not of the object.**

This is §18.4.1 arriving at this paper's expense. A zero is worth the size of the box it was computed in — and an arrow is worth the size of the assumption it was computed under. §7's era structure, §8's reachability figures, §9's two recovery routes, §34's groupoid and Appendix C's O1 are all results about a category whose destinations are empty. **They stand as mathematics. Their reading as cascades does not**, and this paper has called them transitions throughout.

What survives untouched. The occupancy law as mathematics — cap-free, one line, shown able to fail — holds of any index of its stated form, and the extended index simply is not of that form. Nothing in Parts III through VIII runs through composition: the bridge, the n-body index, the pushback floor, the step law, the cut law and the two-index construction are all independent of it.

And the direction, where the clock does hold, is past. Composition never raises k , so a chain runs toward the state that has spent something, and reading it backwards reads toward the fuller, earlier one. §20.1.2 says the same inferentially — the bracket interpolates and does not extrapolate, seven interior cells held against 2.72% and no bound outward — and §18.6 gives the reason, that forward statement is prediction and the budget is $E(X)$. **An index can say where something must have been between two observations; it cannot say where it will be after the last one.**

Established, and about indices. An index whose cells are transitions, *and whose destinations begin empty*, has a temporal structure; that structure is a category; it carries a global time function forced by counting alone; its closed timelike curves exist and are confined to the level sets of that function; every step of one is conservative; and none can host a contradiction, because the language has no negation.

Not established, and not reachable this way. That any of it *is* time. Calling the composition order time is register 313's naming, not a measurement, and §16.7.3 is the warning — a complete consistent index of a city that does not exist certifies itself perfectly. **This section adds a second wall inside the first:** even granting the antecedent, the arrow was carried by an assumption rather than by the structure. The gap was not only between the index and the world.

PART III — THE BRIDGE, AND THE PRESENTATIONS

12. q is a bridge, not an amount

§12.11.5 uses the word: the tight two-parent K is *the first bridge between the cylinder's two ends other than q* , and it breaks the factorisation by 2,475 cells, 3.5% of the product, at the book's caps. Figure 7.4 draws it dashed and forbidden.

The structural content of q was never stated. Computed on the constraint graph $n - \ell - k - q - g - f - e$ with 2S pendant at k :

remove	components	separates the two ends
ℓ	$\{n\} \cdot \text{rest}$	no
k	$\{\ell, n\} \cdot \{e, f, g, q\} \cdot \{2S\}$	no
q	$\{n, \ell, k, 2S\} \cdot \{e, f, g\}$	yes
g	$\{2S, k, \ell, n, q\} \cdot \{e, f\}$	no
f	$\text{rest} \cdot \{e\}$	no

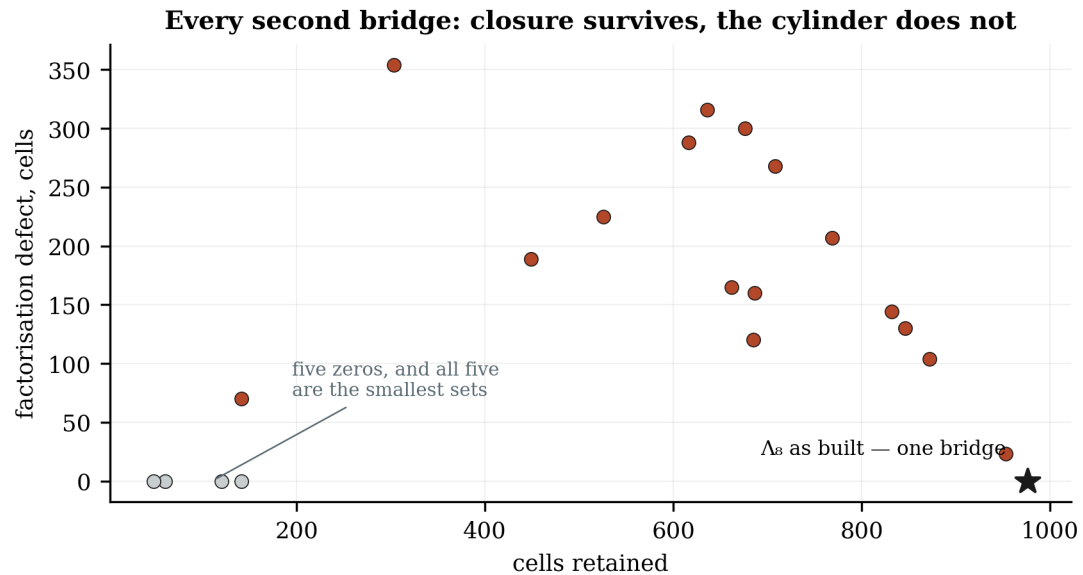
Five cut vertices; **q is the unique one separating source from target.** §13.2's *they meet at q* and §12.6.1's *q playing the role of the conserved coupling* are two readings of one fact the book never states structurally: q is the articulation vertex. Its content as a coordinate is a count; its content as structure is the connection.

13. Every second bridge, priced

Twenty admissible A-side/B-side couplings were imposed on Λ_8 and measured.

second bridge	cells	join fail	meet fail	factorisation defect
<i>none</i> — Λ_8 as built	976	0	0	0
$g \leq n$	953	0	0	23
$2S \leq e$	872	0	0	104
$f \leq 2S$	846	0	0	130
$f \leq \ell$	832	0	0	144
$g \leq 2S$	769	0	0	207
$e \leq n$	708	0	0	268
$\ell \leq g$	686	0	0	160
$g \leq \ell$	685	0	0	120
$e \leq k$	676	0	0	300
$k \leq e$	662	0	0	165
$n \leq e$	636	0	0	316
$\ell \leq f$	616	0	0	288
$2S \leq g$	526	0	0	225
$2S \leq f$	449	0	0	189
$e \leq 2S$	303	0	0	354
$n \leq g$	141	0	0	70
$k \leq g$	141	0	0	0
$e \leq \ell$	120	0	0	0
$k \leq f$	60	0	0	0
$n \leq f$	48	0	0	0

Closure survives every one, as §14.4 guarantees for monotone single-coordinate bounds. **The cylinder survives fifteen of twenty**, with defect 23 to 354. So the exclusivity is exclusivity of the *tree*, not of closure — $E = 0$ is untouched by a second bridge and the factorisation is destroyed by it. §12.11.5's sentence, measured across the whole candidate set rather than at one axis: **the tree is the cylinder, and a bridge is the move that spends it.**



Caution on the five zeros.

Figure 5. Every admissible second bridge between the cylinder's two ends, priced. All twenty give zero join and zero meet failures — §14.4 guarantees that for monotone single-coordinate bounds — and fifteen destroy the factorisation, defect 23 to 354. The star is Λ_8 as built, with one bridge and defect zero. The five pale points at defect zero are the five smallest sets, where a set cut hard enough factorises trivially. The five defect-zero bridges are the five smallest sets, 48 to 141 cells. A set cut hard enough factorises trivially. Register 333's point applies: every closure test in this book passes on the empty index, so a zero is worth the size of the object it was computed in.

14. The mechanism, and it is eight-dimensional

The governing condition is the spanning-tree identity $|E| = |V| - 1$. Λ_8 has eight nodes and seven edges. **To add a bridge and remain a tree you must break one.** That is a statement about the constraint graph, an eight-node object, and it is the whole mechanism.

The book computes three instances and never groups them:

instance	what it buys	what it costs
Λ_9' , the f-g-2S' cycle	93 cells of tightness	Chapter 15's propagation guarantee — <i>the tree or the tightness</i>
Λ_{12} , the tight two-parent K	the law rather than its weakening	2,475 cells at the book's caps, 3.5% of the factorisation
§17.4, conservation $g_1 + g_2 \leq q$	physical correctness	89,864 join failures

And §17.4 is also the escape, the only one. A sum bound cannot be imposed, but a sum can be *indexed*: replace (g_1, g_2) by (g_1, G) with $G = g_1 + g_2$, and conservation holds identically by construction. That converts a forbidden edge into a new **node**, and a tree on nine nodes carries eight edges.

Two moves, and only two: break an edge to make an edge, or add a coordinate to buy one. The second is P16, and §17.4 states its price — coverage and closure together are available; coverage, closure and the original order are not.

By §18.4.1's three-body reading, a second bridge is the three-body problem arriving inside the index: two bodies give one edge and a tree, three give a triangle at treewidth 2.

PART IV — THE n-BODY INDEX

15. What was asked, and what was refused

§18.4.1 identifies the three-body problem's shape — K_3 , treewidth 2, carrying all three of §12.11.2's excluded forms at once, a sum in the superposition, a difference in the relative position, a symmetric function in the mutual interaction. Register 317 records the verdict in one line: **nothing here solves anything.**

That verdict is not disturbed by anything below. Three walls stand and none was moved: Theorem 11.2, that no ordinal structure over these coordinates carries what the coordinates do not, whatever formula the join takes; §18.6, that saying where a body *will* be is prediction and the budget is $E(X)$; and §13.4, that a phase space with the Hamiltonian removed can bound and refuse and cannot predict.

What was attempted instead is narrower and it succeeded: **can the configuration space be indexed and closed?**

16. The filter, extended to four grids

§18.4.1 computes the inverse problem — masses from observation — on one grid. Run at four:

observable	worst-case fibre, n=8	n=12	n=16	n=20
total mass alone	15	21	27	33
total and the largest	4	6	8	10
three pairwise sums	1	1	1	1
one mass ratio	42	78	128	190
two mass ratios	8	12	16	20

The book's $n = 12$ column reproduces exactly — 21, 6, 1, 78, 12. Summed observables degrade linearly in grid size; three pairwise sums separate exactly at every size; **a single ratio is the worst observable available**, because every ratio class is a ray — shape without scale — so no number of ratios pins a mass. This is knowable before any fitting, which is all that transfers: a bracket on a parameter is valid only where the observable is monotone in it, the same test that refused at $\text{Be} \rightarrow \text{B}$.

17. Three presentations, and they are not equivalent

Following §17.4's repair literally: a sum cannot be bounded, so index it.

System A — (m_1, S, M) with $S = m_1 + m_2$ and M the total, recovering $m_2 = S - m_1$ and $m_3 = M - S$. Ordering $m_1 \geq m_2$ becomes $S \leq 2m_1$, one parent. Ordering $m_2 \geq m_3$ becomes $M \leq 2S - m_1$, **two parents** — the tight-K situation exactly — so the book's own move is to weaken it and take the envelope.

System B — (S_{12}, S_{13}, S_{23}) , the three pairwise sums.

System P1 — (m_1, m_2, m_3) , the masses themselves.

Measured at four caps:

cap N	sys	envelope	E(env)	join/meet fail	exact	E(exact)	join fail	meet fail	density
8	A	69	0	0 / 0	41	12	32	0	59.4%
8	B	125	0	0 / 0	41	71	249	229	32.8%
12	A	184	0	0 / 0	102	40	224	0	55.4%
12	B	343	0	0 / 0	102	217	1,633	1,525	29.7%
16	A	385	0	0 / 0	204	90	896	0	53.0%
16	B	729	0	0 / 0	204	464	6,667	6,302	28.0%
20	A	696	0	0 / 0	358	168	2,772	0	51.4%
20	B	1,331	0	0 / 0	358	882	20,757	19,741	26.9%

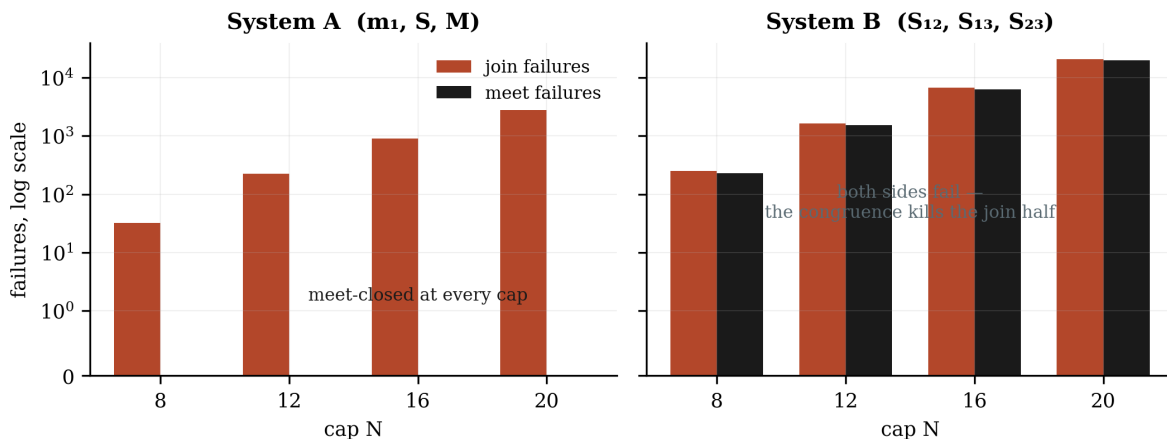


Figure 6. The two presentations fail on different sides. System A's exact set is meet-closed at every cap — zero, at all four — because its ordering bound is decreasing in m_1 and a decreasing cap breaks joins, which §18.4.1 computed on chemistry before this object existed. System B fails both, and §18 isolates why: the parity congruence kills the join half that A.18 would otherwise have supplied.

Both envelopes are closed indices, $E = 0$ at four caps each. System A's exact set is **meet-closed** at every cap — zero meet failures. System B fails both sides.

The failure direction was predicted by §18.4.1's cap table. $M \leq 2S - m_1$ is **decreasing in m_1** , and the book's chemistry computation gives $\varphi(a) = 4$ — a with 875 join failures and 0 meet failures — the reason coordination number and lone pairs resist indexing while oxidation state and bond order do not. The three-body ordering constraint is a decreasing cap and fails the way decreasing caps fail. That is §2.17's triangulation: a criterion measured on chemistry predicting a failure direction in celestial mechanics.

System B's gap decomposes into three machines, mirroring §12.11.2:

cap N	triangle	congruence	mass cap	total
8	30	40	14	84
12	91	112	38	241
16	204	240	81	525
20	385	440	148	973

The three parts sum exactly at every cap.

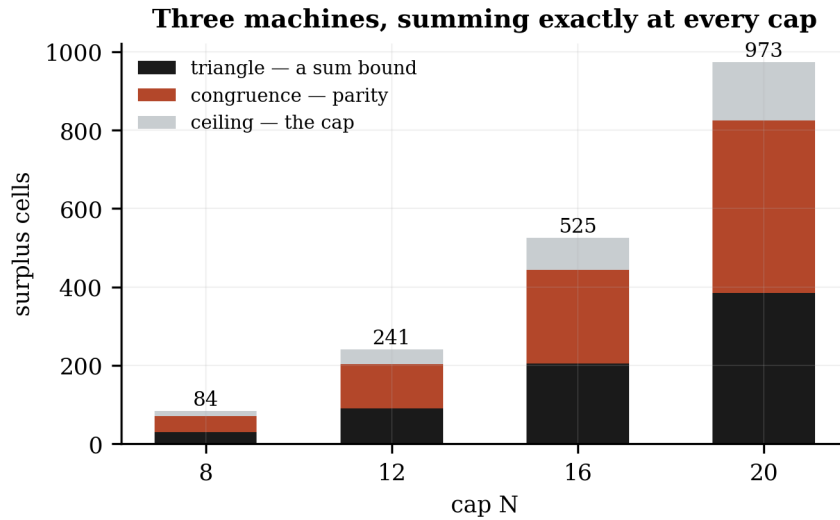


Figure 7. System B's envelope surplus, decomposed. Three machines — a sum bound, a congruence, a ceiling — summing exactly at every cap. This is §12.11.2's decomposition of the tower's envelope gap, arriving in a subject with no angular momentum in it, which is the first evidence that the counting/coupling dichotomy is about indexing rather than about coupling. §12.11.2 decomposes the tower's envelope gap into reflection, congruence and triangle; here it is ceiling, congruence and triangle — **the same shape in a subject with no angular momentum in it.**

18. A failed prediction, converted to a bound

Committed: system B would fail meets only, by A.18, since the triangle region is join-closed and meet-broken.

Refuted: it fails both. Isolated:

region	join fail (cap 20)	meet fail (cap 20)
pure triangle $\{c \leq b \leq a \leq b+c\}$	0	32,890
triangle + parity congruence	26,328	30,745

A.18 transports exactly — the parity-free triangle is join-closed at every cap, 0 failures at 8, 12, 16 and 20. **The congruence kills the join half**, and §12.11.2 states this outright: *impose the parity congruence as well and the join closure dies too, so the surviving half belongs to the parity-free triangle and to nothing larger.* The prediction applied A.18 to a region carrying a congruence, which the book had already excluded.

Per §2.15, the exclusion is the product:

A.18's join-closed half cannot be transported to any integer-valued system, because integers force parity. The surviving half is unusable wherever the quantities are counts — which is every index in *The Method*.

19. The triangle of presentations, and the Ravi identification

Writing B's coordinates against A's: $a = S_{12}$, $b = S_{13}$, $c = S_{23}$, so $a + b + c = 2M$ and

$$s - a = m_3, \quad s - b = m_2, \quad s - c = m_1$$

The masses are the tangent lengths of a triangle whose sides are the pairwise sums, and the total mass is the semiperimeter. This is the Ravi substitution, and every computed result falls out of it: non-negativity $m_3 \geq 0$ is the triangle inequality; mass ordering is side ordering; and the parity congruence is the condition that integer sides give integer tangent lengths — a real geometric condition, not an artefact of encoding.

The three presentations are then a conservation of difficulty:

presentation	carries	ordering	non-negativity	total-mass cap
P1 tangent lengths	no observable	free	free	inexpressible
A one mass, two sums	interior	envelope	free	free
B three sides	every observable	free	envelope	free

Each vertex is free in exactly two of three and pays in the third. **No vertex is free in all three, and the payment relocates rather than vanishing.** P1 closes perfectly — 165, 455, 969, 1,771 cells at caps 8, 12, 16, 20, $E = 0$, zero join and meet failures, density 100% — and cannot state a total-mass cap at all, because the cap is a sum. It indexes what cannot be observed.

That is §6.3's calendar in a second subject: $E = 7$ is the price of keeping January first, and the envelope gap is the price of carrying a summed observable. **Non-closure as a purchase, not a fault.**

One coincidence, tested and correctly read. At $N = 16$, $|A \text{ exact}| = 204$ and B's triangle gap = 204. Across four caps: 41 vs 30, 102 vs 91, 204 vs 204, 358 vs 385. There is no bijection — that reading is refuted. But two monotone integer sequences of like degree with different leading coefficients cross exactly once, so **the crossing is forced and $N = 16$ is where.** The first draft of this analysis called it a coincidence; in a closed index there are none, and the correct statement is a single determined crossing.

20. The elevation, and what survives it

B does not elevate. At $n = 3$ there are 3 masses and 3 pairwise sums, exactly determined. At $n = 4$ there are 4 masses and 6 pairwise sums, over-determined by two. The sides-presentation exists only at three bodies, which is why that vertex is a triangle and cannot become a tetrahedron.

A elevates, as a chain. Partial sums always give n coordinates for n masses. Since $m_{\{k+1\}} \leq S_k / k$, the envelope is

$$S_1 \leq S_2 \leq \dots \leq S_n, \quad S_{k+1} \leq \frac{k+1}{k} S_k$$

single-parent and monotone at every k — a **chain of Cesàro bounds**, treewidth 1 at every n .

Which also corrected the construction. System A used $M \leq 2S$ where the tight bound is $M \leq (3/2)S$. Every density reported above understates by roughly a third:

cap N	loose envelope	density	tight envelope	density
8	69	59.4%	53	77.4%
12	184	55.4%	142	71.8%
16	385	53.0%	294	69.4%
20	696	51.4%	526	68.1%

Same class as §12.11.1's Λ_9' : an admissible tightening left on the table because the loose bound closed and nobody asked whether a tighter one also did.

The n-body table.

n	N	envelope	E	join fail	meet fail	exact	density
3	12	142	0	0	0	102	71.8%
3	16	294	0	0	0	204	69.4%
3	20	526	0	0	0	358	68.1%
4	12	341	0	0	0	155	45.5%
4	16	876	0	0	0	359	41.0%
4	20	1,876	0	0	0	717	38.2%
5	12	712	0	0	0	197	27.7%
5	16	2,248	0	0	0	509	22.6%
5	20	5,668	0	0	0	1,125	19.8%

Exact-set join failures run 224 / 896 / 2,772 at $n = 3$, 1,073 / 5,912 / 23,904 at $n = 4$, and 2,491 / 17,448 / 87,398 at $n = 5$. **Meet failures are zero in all nine.**

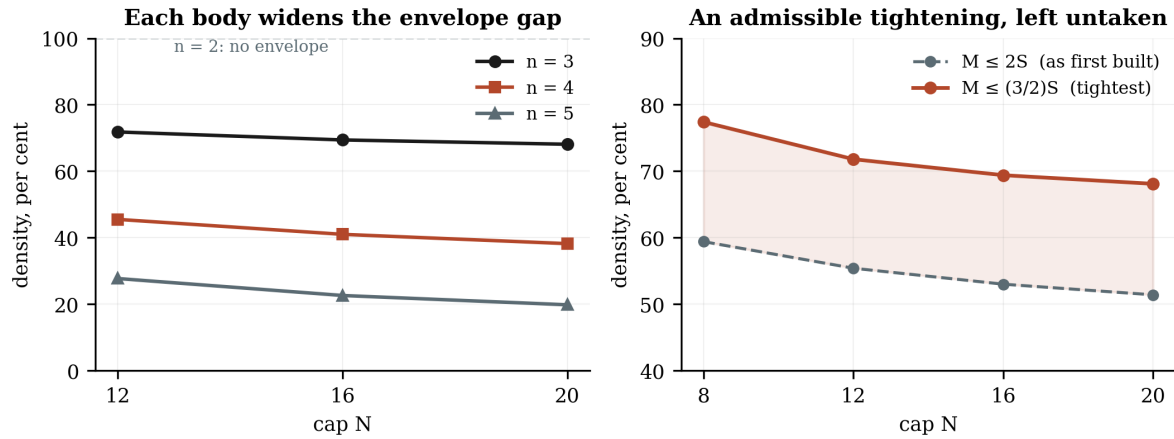


Figure 8. Left: each added body widens the envelope gap — 68.1%, 38.2%, 19.8% at $N = 20$ — which is §12.11.0.2's fill law in a second subject. Two bodies give a tree with no envelope at all, so three bodies is the second term of a sequence rather than a threshold. Right: the shaded band is register 6 of Part VI, an admissible tightening left untaken. Every density first reported understated by about a third.

Three bodies is not a threshold. It is the second term of a sequence. Two bodies give a tree with no envelope at all; three cost 32%; four cost 62%; five cost 80%. Each added body multiplies the ambient chain faster than it multiplies the object, which is §12.11.0.2's fill law in a subject with no shells in it.

21. Meet-closure, proved

Rewriting the coordinates as the sequence itself: S is the partial-sum vector of a non-increasing non-negative m , which says exactly that S is **concave** — $S_{\{k-1\}} + S_{\{k+1\}} \leq 2S_k$. The envelope condition $S_{\{k+1\}} \leq ((k+1)/k)S_k$ says exactly that S_k/k is **non-increasing** — the Cesàro means fall. Concavity implies falling means; the converse fails, and that failure is the density gap.

Proposition. The exact set is closed under componentwise minimum.

Proof. Let $f = \min(g, h)$ pointwise with g, h concave, and suppose $f(k) = g(k)$. Then $f(k-1) \leq g(k-1)$ and $f(k+1) \leq g(k+1)$, so $f(k-1) + f(k+1) \leq g(k-1) + g(k+1) \leq 2g(k) = 2f(k)$. Hence f is concave. Non-negativity, monotonicity and the cap survive a minimum trivially. ■

The same argument on max reverses at the first step — $f(k-1) \geq g(k-1)$ is the wrong direction — and no repair is available. **The minimum of concave functions is concave; the maximum is not.** That is the whole of the one-sidedness, at every n and every cap, without further computation.

And it settles the parity question of §18. §12.11.2's triangle is a *sum* bound and survives the join; concavity is a *difference* bound — a second difference — and survives the meet. A.18's remark says max moves a coordinate toward the side each bound is slack on; second differences are slack downward, so the surviving operation flips. **The two excluded forms have opposite survivors, for a stateable reason.**

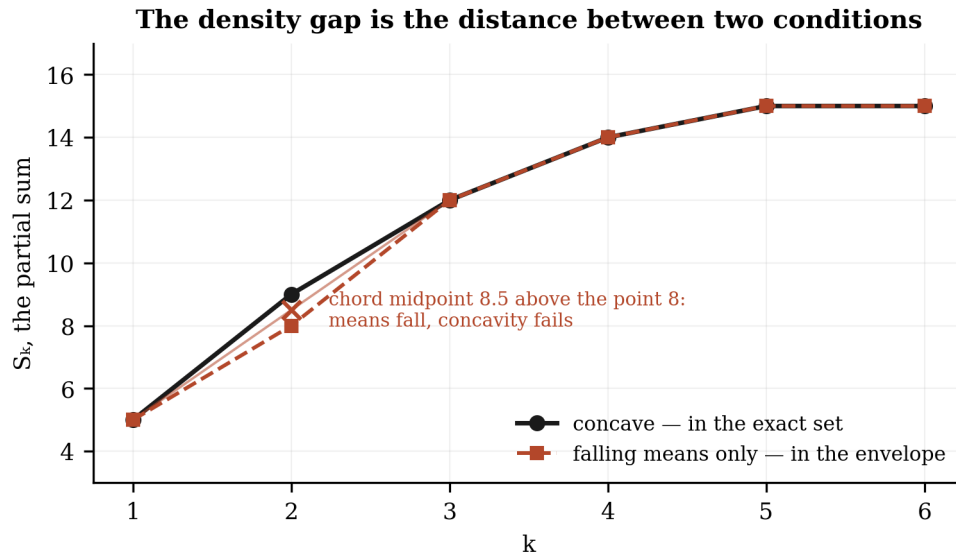


Figure 9. What the density measures. The exact set is the concave sequences; the envelope is those whose Cesàro means fall. Concavity implies falling means and the converse fails, and every cell of the gap is a sequence like the dashed one — means falling, a chord rising above its own point. $E(X)$ counts them.

22. $\mathcal{R}$ answers the tightness question

Committed: $\mathcal{R}(\text{exact})$ would be strictly smaller than the hand-built chain at $n \geq 4$, since the chain imposes only adjacent bounds while $\mathcal{R}$ recovers every pair, and §D.4.3 reports the recovered system carrying bounds on all coordinates with the binding one changing cell by cell.

Refuted, and the refutation is the stronger result.

n	N	exact	chain envelope	$\mathcal{R}(\text{exact})$	$E(\text{exact})$	tightest?
3	12	102	142	142	40	yes
3	16	204	294	294	90	yes
3	20	358	526	526	168	yes
4	12	155	341	341	186	yes
4	16	359	876	876	517	yes
4	20	717	1,876	1,876	1,159	yes
5	12	197	712	712	515	yes
5	16	509	2,248	2,248	1,739	yes
5	20	1,125	5,668	5,668	4,543	yes

$\mathcal{R}(\text{exact})$ equals the chain envelope as sets, nine of nine. The path of adjacent bounds is transitively closed — composing them gives $S_k \leq k \cdot S_1$, which is exactly the direct bound $\mathcal{R}$ recovers — so non-adjacent pairs carry nothing new. Verified directly at $n = 4$, $N = 16$: the two sets are equal, $\mathcal{R}$ is idempotent on it, and join and meet failures are both zero.

This also fills the cell that timed out: **$E(\text{chain}) = 0$ at $n = 5$** follows without separate computation.

Unlike Λ , this object's **defining system is its own normal form**. §D.4.4 records that Λ 's defining system carries seven bounds while its normal form has sixteen constraining of fifty-six recovered, and the map from presentation to normal form runs one way. Here the map is an identity, and by register 325's formulation that is $E(X) = 0$ read as *the extent is the law for that set*.

PART V — PAST, PRESENT, FUTURE

23. The balance, and where it holds

§10.2 reads the constraint tree as *past — change — future*, left to right: the source end (n, ℓ , k, 2S), the transfer q, the target end (e, f, g). The three-part reading is the book's; what follows is its arithmetic.

The three parts do not add up as three free things.

$$|A| \cdot |Q| \cdot |B| = 33 \cdot 4 \cdot 17 = 2,244 \quad \text{against} \quad |\Lambda| = 976$$

A defect of **1,268** — the three parts overcount by 130% when treated as independent. Conditioned on the transfer they separate exactly: $\sum_q |A(q)| \cdot |B(q)| = 976$, defect zero.

And the balance is not special to the transfer. Tested at every cut of the constraint tree, four of the eight coordinates split it in two — ℓ , q, f and g — and **all four give defect zero**. §12.6.1 presents the q-factorisation as though the transfer were doing the work; the computation says the **tree** is doing it, and q is distinguished only by *which* two things it separates.

The shape is a path, not a triangle. Reading the three parts off the constraint tree gives edges $\text{past} \leftrightarrow \text{present}$ and $\text{present} \leftrightarrow \text{future}$, and **no past–future edge**. Two edges, treewidth 1. That is not decoration: §18.4.1's criterion is that treewidth 1 closes exactly and treewidth 2 gives an envelope, so **Λ 's exactness is a proof that the three parts are not a triangle**. A triangle would be the three-body shape, and register 316 states its price.

24. The future is a bracket, and the past does not enter

Two questions with computable answers: is the future a function of past and present, and does the past enter at all?

(past, present) pairs	97
futures per pair	min 5 · median 10 · max 17
pairs determining a single future	0 of 97

Not one pair determines a future. That is §20.1's bracket arriving at the level of the transition itself, and §13.4's line computed — structure does not choose, and 1,113,045,672 maximal chains satisfy every constraint in the book while it chooses none of them.

And for each value of the present, all pasts give the same future set.

present q	distinct pasts	distinct future sets	future
0	33	1	5
1	33	1	10
2	23	1	15
3	8	1	17

The future is conditionally independent of the past given the present. A Markov property, exact at every value.

That is the arithmetic behind §23's balance: the factorisation *is* the conditional independence, and the 1,268 is what treating the three parts as free costs. The past constrains only which presents are reachable — $q \leq k$ — and having spent that, it has spent everything.

The dependence is monotone and opposed. The future set is **nested increasing** in the present — $5 \subset 10 \subset 15 \subset 17$, each containing the last — while the past set is **nested decreasing** — $33 \supseteq 33 \supseteq 23 \supseteq 8$. §12.8.1's Pareto frontier, and the same shape as $dw/dh > 0$ with $dV/dh < 0$.

And the bracket is saturated. At every value of the present the admissible future set is the *full* interval of the target lattice between its own extremes, $[(1,0,0), (3,1,q)]$ — nothing inside is missing. $E(X) = 0$ read on the target end.

25. The tick, decomposed

The clock is occupancy and the tick is $k - g$. Over the 1,169 composable cells of Λ_9 :

tick	cells	share
0 — reversible	389	33.3%
1	540	46.2%
2	240	20.5%

The tick has two sources and they are exactly balanced. In universal form, for a cell a with weight w on objects and transfer t on cells,

$$w(\text{src } a) - w(\text{tgt } a) = \underbrace{[w(\text{src } a) - t(a)]}_{\text{retained: never transferred}} + \underbrace{[t(a) - w(\text{tgt } a)]}_{\text{transferred but not received}}$$

and both brackets are non-negative exactly under the transfer bound $t(a) \leq w(\text{src } a)$ and non-accretivity $w(\text{tgt } a) \leq t(a) + w_0(a)$ with $w_0 \equiv 0$. **The clock's tick is retention plus loss in transit, and neither term is privileged.** In Λ_9 's coordinates this reads $k - g = (k - q) + (q - g)$: something not removed, and something removed but not placed.

at the caps of §7.4	cells	share
something not removed, $k > q$	430	36.8%
something removed and not placed, $q > g$	430	36.8%

Not approximately — 430 against 430, with the split table its own transpose: $(0,1) = 270$ against $(1,0) = 270$, and $(0,2) = 80$ against $(2,0) = 80$. **Irreversibility has two independent sources contributing identically.** One is a choice not taken; the other is a capacity that could not receive.

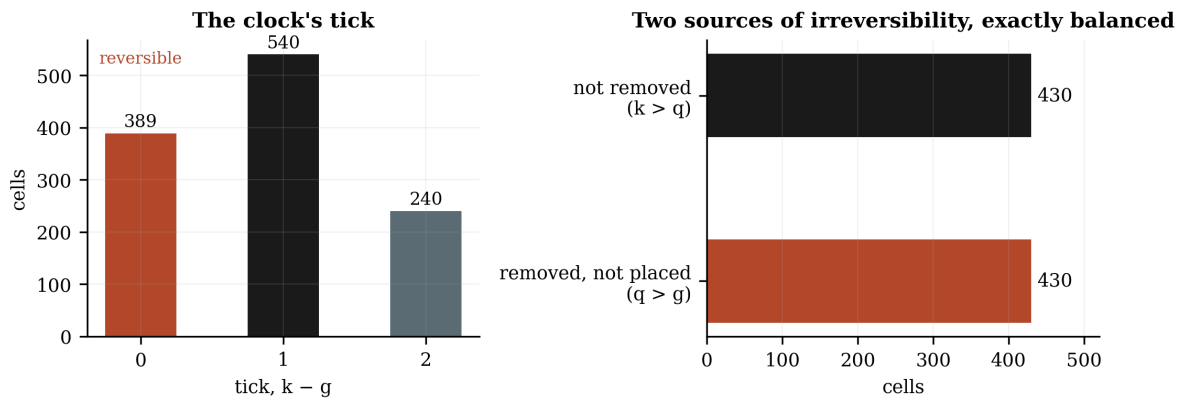


Figure 10. The clock ticks 0, 1 or 2, and the two ways it ticks are exactly balanced — 430 against 430, with the split table its own transpose. One source is a choice not taken, the other a capacity that could not receive, and the index weights them identically.

And Pauli is a minority partner in the tick. Over the composable cells of Λ_9 at the caps of §7.4, g 's ceiling is fixed by counting in 60.2% of composable cells, tied in 29.5%, and by the shell capacity in **10.3%** — with only 6.2% where Pauli holds g down and the transfer would have allowed more. **The arrow of time in this object is overwhelmingly a counting phenomenon**, which §12.9 states in advance, at the same caps: the coupling $g \leq q$ binds 35.6% of intervals and $g \leq 4f+2$ binds 4.9%.

PART VI — WHAT CANNOT BE REMOVED

26. Pushback

For a cell x , count the pairs of *other* cells that produce it — $J(x)$ joining to x , $M(x)$ meeting at x . Remove x and that is exactly how many pairs of survivors fail.

minimum over all 976 cells	16
median	506
maximum	5,091
mean	739
cells removable with no failing pair	0

No cell of Λ_8 can be removed without the remainder contradicting the removal. The weakest is implied back sixteen ways.

The formula is exact rather than fitted. $J(x)$ recomputed using only the cells at or below x agrees with the global count on 930 of 930 cells checked; $M(x)$ inside $[x, T]$ on 934 of 934. So $J(x)$ is the number of ways x 's own down-set is covered by two proper down-subsets, and $M(x)$ the dual: **pushback is a property of the interval and of nothing outside it.**

Which is why statistics fail — $\text{corr}(\text{pushback}, \text{cells below}) = +0.425$, cells above $+0.454$, their product $+0.117$. §16.8.4 found precisely this for insertion: four statistical measures of a constraint all failed and only a measure of the *tree* succeeded. Same shape, second operation.

$J(x) = 0$ exactly on the join-irreducibles and $M(x) = 0$ exactly on the meet-irreducibles, both exactly, 17 each with $\perp$ and $\top$ excluded — reproducing §8.3's count by a route with nothing to do with Birkhoff.

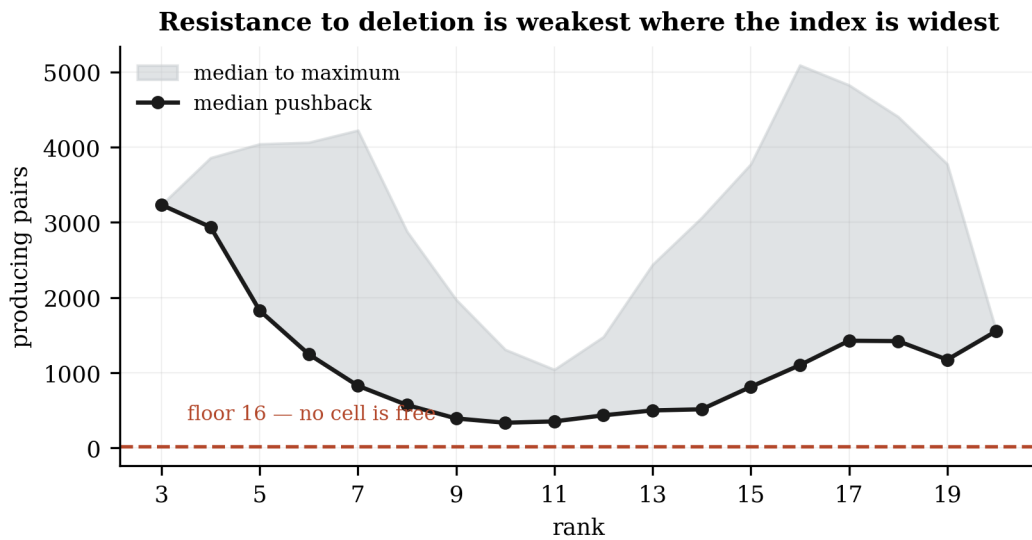


Figure 11. How hard the remainder pushes back against losing each cell, by rank. The floor is 16 and nothing reaches zero, because zero would require a cell irreducible on both sides and Λ has none. The extremes are defended hardest; rank 11, which holds the most cells, defends each of them least.

And the distribution is U-shaped in rank. Median pushback runs 3,234 at rank 3, down to 335 at rank 10, back up to 1,555 at rank 20. **An index's resistance to deletion is weakest where it is widest** — rank 11 holds 122 cells and defends each of them least — which is the opposite of the intuition that more neighbours means more redundancy.

27. The step law

A single cell cannot leave. The minimum that can is derived rather than measured.

Derivation. Let $R = [a, b]$ and suppose $z \in R$ with $z = xvy$ and $x, y \notin R$. Since $x, y \leq z \leq b$, being outside R means $x \not\geq a$ and $y \not\geq a$, while $a \leq xvy$. In a distributive lattice every join-irreducible is join-**prime**, so $a \leq xvy$ forces $a \leq x$ or $a \leq y$ — contradiction. Conversely if a is join-reducible, $a = uvv$ with $u, v < a$, both under b and neither above a , so $u, v \notin R$ while $uvv = a \in R$. Dually for meets and b . ■

$X \setminus [a, b]$ is a sublattice if and only if a is join-prime and b is meet-prime.

$\text{step}(X) = \min \{ |[a, b]| : a \text{ join-prime, } b \text{ meet-prime, } a \leq b, 1 < |[a, b]| < |X| \}$

Corrected by its own test. The first statement of the criterion used *irreducible* rather than *prime* and disagreed with

the truth on **35 of 116,138 intervals** — exactly $17 + 17 + 1$, being the intervals $[\perp, b]$ with b meet-irreducible, $[a, \top]$ with a join-irreducible, and the whole lattice. Convention excludes $\perp$ from the join-irreducibles; the property the proof uses is primeness, which $\perp$ satisfies trivially. **With prime: 0 disagreements in 116,138 intervals**, exhaustive.

caps	$ \Lambda $	$ J = M $	doubly irreducible	step
(3,3,1,3)	976	17	0	4
(3,3,1,4)	1,636	21	0	4
(4,4,1,3)	1,968	19	0	9
(4,4,1,4)	3,363	23	0	9
(5,5,1,3)	3,304	21	0	16

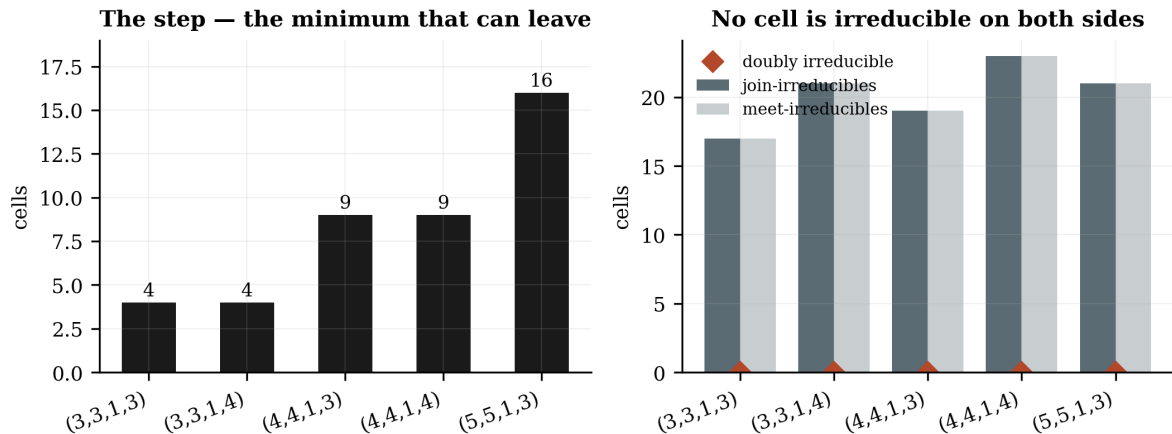


Figure 12. Left: the minimum set that can leave, at five cap settings. Right: join- and meet-irreducibles are equinumerous at every setting and **no cell is both** — which is why the step is never one, and why the pushback floor of §26 is never zero. The two results are one fact.

At the book's caps the step is a coordinate box free in exactly two coordinates — **n and e, the two shell coordinates, the two ends of the caterpillar** — running from $(2,1,3,3,2,1,3,0)$ to $(3,1,3,3,3,1,3,0)$, and removing it leaves 972 cells with **0 join/meet failures across all 471,906 surviving pairs**.

The step is $(n_{\max} - 1)(e_{\max} - 1)$ at all five settings. That is a pattern across five points and it is **not promoted**: §20.1.2 forbids taking the next term from the pattern the terms make. What is derived is the criterion; a closed form for the minimum over primes is not.

28. No cell is doubly irreducible

The two results of §26 and §27 are one fact seen twice.

$[a,a] = 1$ requires a cell that is both join-prime and meet-prime. **Λ has none, at every cap tested** — 17 join-irreducibles, 17 meet-irreducibles, intersection empty. So the pushback floor of 16 and the step of 4 are the same statement: there is no cell whose removal the rest of the index does not contradict, because there is no cell that is irreducible on both sides.

And the E1 puncture is placed by it. §12.11.6 removes the one cell nature forbids from the E1 selection rule and reports closure failing by exactly four meets, all four producing the removed cell. Recomputed at cap 20: **0 join failures, 4 meet failures, all four producing (0,0)**. Four is not typical — it is near the floor. **The forbidden cell is among the least-defended its index has, and the index still refuses to lose it.**

PART VII — THE ONE COUPLING

29. The cut law

Opposed monotonicity — two quantities moving oppositely under one free parameter — appears at least nine times in *The Method 1.3* and this session, and is never grouped: §12.8.1's Pareto frontier, §21.5.2's dw/dh against dV/dh ,

§12.11.1's *tree or tightness*, §12.11.0.1's fill against surplus, §6.3's calendar, §17.4's coverage against order, §21.10.3's precision against robustness, §29.3.3's Hodge slice, §30.4.1's paradox-freedom against expressive power, and §24 above.

The lattice-internal instances are derivable.

At a cut of the constraint graph on coordinate v , each surviving component shrinks if it contains a parent of v , grows if it contains a child of v , and is flat if it contains neither. Raising v raises the floor on its parents and the ceiling on its children.

Tested across all seven stages of the tower and every cut: 89 component tests, one violation. And the violation is at Λ_9' , whose constraint graph is the only one in the tower carrying a cycle — $f-g-2S'$, the tightening §12.11.1 prices at 93 cells and calls *the tree or the tightness*. The law is a theorem about trees and fails at exactly the stage that stopped being one.

A prior version of this claim was wrong and is withdrawn. It read *opposition fails at two-parent coordinates*, which holds at Λ_8 and nowhere above. The cause is not two parents: at Λ_8 the coordinate g has two parents **and no child**, so every component holds a parent and every one shrinks. From Λ_9 onward g has the child $2S'$ and opposition returns.

30. The two-parent coordinate at every stage of the tower

§11.5 names $g \leq \min(q, 4f+2)$ *the one coupling, and it is the Pauli principle* — the only non-product term in the whole expression. The tower confirms it is the only one anywhere.

stage	X	two-parent coordinates	cut behaviour at g
Λ_8	976	g	all components shrink
Λ_9	1,654	g	opposed
Λ_9'	1,561	g and $2S'$	the one law violation
Λ_{10}	2,535	g	opposed
Λ_{11}	13,585	g	opposed
Λ_{12}	70,905	g	opposed
Λ_{13}	199,130	g	opposed

g is the unique two-parent bound in the entire tower as built. Every coordinate from the ninth axis to the thirteenth enters with exactly one parent, which is why closure survives thirteen dimensions and forty-seven million ambient cells.

And the two places a second could have entered are the two places the book records a price. Λ_9' makes $2S'$ two-parent — 93 cells bought, the tree lost, and the only violation of the cut law in the run. Λ_{12} 's law for K is $2K \leq 2J_c + 2f$ with the cell's own f , and §12.11.3.1 records the book substituting $f_{\max}$ to keep one parent and the cylinder.

31. Every second bridge, at the top of the tower

§17's twenty candidates at Λ_8 are repeated at Λ_{13} with sixty-four.

As built the factorisation over q is exact at Λ_{13} — defect 0 on 199,130 cells, with sections $11,470 \cdot 45,880 \cdot 89,700 \cdot 52,080$ reproducing §12.11.5's printed table to the cell.

Fifty-eight of sixty-four second bridges destroy it, defects from 1,794 to 82,680. The six that do not are the six that reduce the object to a remnant — at the caps of §7.4, $k \leq g$ retains 27.1% of the cells and $k \leq f$ only 2.5%. Register 333's caution: a set cut hard enough factorises trivially.

The bridge is one-dimensional not because the index has one dimension available, but because the physics supplies one demand and the construction refuses the second.

Structurally a second bridge is always admissible — closure survives all sixty-four, as §14.4 guarantees for monotone single-coordinate bounds. Physically the tower offers exactly one: $2K \leq 2J_c + 2f$, core angular momentum to target orbital momentum. Dimensions 1–10 are single-parent laws, dimension 11 takes its bound from k alone, dimension 13 from K alone.

And q is the unique cut vertex separating the two ends. The path from g to K runs $g - q - k - 2J_c - K$, and removing q disconnects them; the cycle the tight K would create is $K - 2J_c - k - q - g - f - K$, length

six. The forbidden bridge does not add a new connection between the ends. It adds a *second* one, which is what a tree forbids.

32. The tight K, and a discrepancy

§12.11.5 states that the tight two-parent K breaks the factorisation by **2,475 cells, 3.5% of the product**. Recomputed here by four disjoint routes, all at the caps of §7.4:

route	Λ_{12}	Λ_{13}
cells cut by restoring the law	15,150	45,450
factorisation defect once restored	15,150	45,450
per-section defect, summed	—	$2,934 + 11,736 + 22,140 + 8,640 = \mathbf{45,450}$
three alternative A/B splits	—	$113,325 \cdot 405,715 \cdot 428,340$

Four routes agree; **none produces 2,475**. What does is arithmetic on the stated percentage: 3.5% of the 70,905 cells at these caps = **2,482**. So the two figures printed side by side appear to be one measurement written twice, which is §2.8's vacuous identity — computed along one path, it cannot disagree with itself.

The rebuild is cleared under §4.1. It reproduces all seven tower cell counts, the four-cap sweep, the composition closure at 41,682 pairs, and §12.11.5's own four sections exactly.

The true fraction is 21.4% at Λ_{12} and 22.8% at Λ_{13} — and it is not constant: at $f_{\max} = 2$ the cost rises to **35.3%**. The price of keeping the tree grows with the shell the index carries, which is §12.11.1's density behaviour in a second guise and which neither the book nor this session had before.

Two possibilities remain and the triangulation does not settle between them: 3.5% may measure something narrower than *the product* that this rebuild has not reconstructed, or the cylinder costs six times what the section states. Recorded as a discrepancy to localise rather than a correction to claim.

PART VIII — TWO INDICES

33. The reversal

Reversing time is not a sign change. Extending k , q and g below zero was tested at three settings — $k \geq 0$, then all three to -1 , then to -2 — and the object stays closed (0 join/meet failures) while **0 cells have $g > k$ and 0 steps raise occupancy**, the lattice growing from 976 to 2,691. Neither §7.3's closure proof nor the occupancy law uses positivity. **A negative occupancy is not a reversed one; it is a lower one.**

The reversal is a different object. Exchange the two ends of Λ_9 — $(n, \ell, k, 2S) \leftrightarrow (e, f, g, 2S')$, transfer unchanged:

$ \Lambda_9 $	1,654
$ \text{rev}(\Lambda_9) $	1,654
$\text{reverse}(\Lambda_9)$ closed	yes , 0 failures
$ \Lambda_9 \cap \text{rev}(\Lambda_9) $	389

The reversed index is a lawful closed index of the same size and it is not Λ_9 . That is §8.4's failure of self-duality read temporally: the two are both lawful and simply not the same object.

34. The intersection is a groupoid

Every cell of the intersection satisfies **$g = q = k$** — total transfer, nothing left behind — and the intersection is the reversible region twice over: it is exactly the 389 morphisms carrying a reverse.

position	cells	join fail	meet fail	$E(X)$
Λ_9 forward	1,654	0	0	0
$\text{rev}(\Lambda_9)$ backward	1,654	0	0	0
$\Lambda_9 \cap \text{rev}(\Lambda_9)$	389	0	0	0

And it is a groupoid, entire. 4,887 composable pairs with zero closure failures; **all 33 objects carry a genuine identity; all 389 morphisms have a two-sided inverse.** Λ_9 is a category and a groupoid only at its top; the intersection *is* the groupoid.

It carries its own transfer and factorises exactly — $\sum_q |A(q)| \cdot |B(q)| = 389$ against 389, defect zero — so the whole construction reproduces one level down.

The second present definable from the forward and backward indices is the region in which time has no direction, and it is 23.5% of Λ_9 at the caps of §7.4.

This unifies two results reported separately. §7's cycles were confined to level sets of occupancy with every step conservative; conservative is $g = q$ and the level-set condition is $k = g$. **The intersection is that condition stated as an object rather than as a property of paths.**

35. The union does not exist

position	cells	join fail	meet fail	E(X)
$\Lambda_9 \cup \text{rev}(\Lambda_9)$	2,919	360,000	470,625	2,857

Three positions exist; the fourth does not

union: 2,919 cells · 830,625 failing pairs · E = 2,857 — not closed

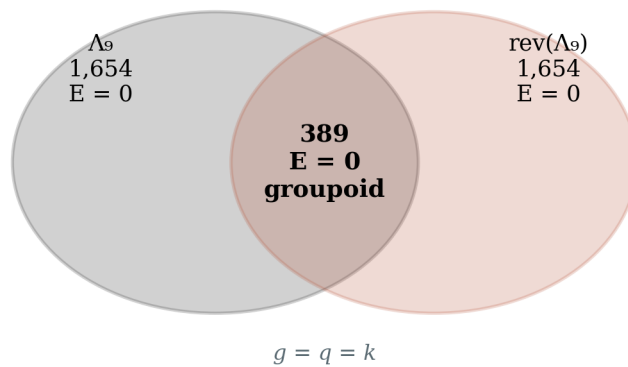


Figure 13. The forward index, the reversed index, and their agreement. Each of the three closes at $E = 0$; the intersection is the region where the transfer is total and nothing is lost, and it is a full groupoid. The union is not a position — 830,625 failing pairs and $E = 2,857$.

Three of the four positions exist and the fourth does not. **The top is missing.** A present is definable from the past and future indices; the thing containing both is not.

That is not a technicality. §17's E1 admits cells when their joins and meets fall back inside; here **2,857 further cells** would have to be admitted, and §16.8.4's amplification is exactly that cost. **A structure containing both directions of time is not available at any price the index recognises** — only a structure containing their agreement.

PART IX — THE ARROWS

36. One clock per coordinate

§11 shows the arrow does not survive an index that can represent occupied destinations: in the 13,775-cell extended object the composition graph is a **single strongly connected component** over all 33 objects, and every one of the ten coordinates rises and falls. No monotone weight exists, and none can — a monotone weight gives a topological order and a strongly connected graph admits none.

What survives is a sub-index, and it is the largest one. The condition is not that destinations be empty but that they be **non-accretive**:

Definition. For a weight w on objects and a transfer t on cells with $t(a) \leq w(\text{src } a)$, decompose $w(\text{tgt } a) = w_0(a) + t(a)$ into prior content and arrival. The index is **non-accretive** when $w_0(a) \leq w(\text{src } a) - t(a)$ — *the destination's prior content never exceeds what the source retains* — equivalently $w(\text{tgt } a) \leq w(\text{src } a)$.

Theorem. The maximal sub-index on which w is non-increasing along composition is $X_w = \{a : w(\text{tgt } a) \leq w(\text{src } a)\}$. It is the *maximum*, not merely maximal: any X carrying the clock satisfies the condition cellwise, so $X \subseteq X_w$, and X_w is itself closed.

index	cells	pairs	join	meet	steps raising w	eras
extended, accretion free	13,775	94,868,425	0	0	826,950 = 31.6%	1
extended + $w(\text{tgt}) \leq w(\text{src})$	5,165	13,336,030	0	0	0	$3 \cdot 15 \cdot 10^8$
Λ_9 as built, $w_0 \equiv 0$	1,654	1,367,331	0	0	0	$3 \cdot 15 \cdot 10^8$

Empty destinations are **sufficient and not necessary**, and the clock survives on an object **three times the size** of the one the book built. Λ_9 is a proper sub-index of the maximal one.

37. Four arrows, and only four

Nothing in the definition mentions occupancy. Every weight gives a candidate arrow, and the admissible ones are exactly the coordinates.

weight	cells	join	meet	rises	eras	admissible
shell	9,407	0	0	0	$3 \cdot 14 \cdot 14 \cdot 5$	yes
subshell	9,845	0	0	0	$2 \cdot 18 \cdot 15$	yes
occupancy	5,165	0	0	0	$3 \cdot 15 \cdot 10^8$	yes
spin	7,160	0	0	0	$4 \cdot 12 \cdot 12 \cdot 7 \cdot 2$	yes
shell + occupancy	4,897	579,460	171,552	0	5	no
rank, all four	4,817	698,337	227,243	0	9	no
max(shell, occupancy)	5,151	0	578,416	0	3	no
min(shell, occupancy)	8,623	2,681,376	0	0	3	no

Sums break both sides, max breaks meets, min breaks joins — §18.4.1's cap table deciding once again what can be a clock. And the four arrows are genuinely different objects: three eras, two, three, four, with different sizes, and occupancy — the one the book built on — the **most restrictive** of the four.

Proved, and the premise shown to be what does the work. X_w is the constraint $x_j \leq x_i$, a bound of one coordinate by the **identity**, so §7.3's argument gives closure directly. Substituting a decreasing φ breaks 678,750 joins and a non-monotone φ breaks 67,500.

And each coordinate carries a trichotomy. The **co-arrow** $w(\text{tgt}) \geq w(\text{src})$ is the same form of bound and also closes — $10,103 \cdot 11,933 \cdot 10,900 \cdot 9,790$ cells, and **not one composable step lets the weight fall** across 1.18, 1.68, 1.40 and 1.04 million steps. Arrow and co-arrow **cover** the object, since $\leq$ is total, and meet in the **constant core** — $5,735 \cdot 8,003 \cdot 2,290 \cdot 3,175$ cells, closed. Occupancy's core is the smallest, which is why its arrow is the strictest.

38. The arrows form a meet-semilattice

Intersections of arrows are closed — an intersection of sublattices is a sublattice, one line, no caps — and the sixteen combinations are **all distinct**. Unions never close: all six fail, and the generated joins are proper at four cap settings, twenty-four joins, worst case 96.5% of the ambient.

But a join of two arrows is not an arrow. No join size coincides with any arrow size, six of six. So the arrows are a **meet-semilattice inside the complete lattice of sublattices**: closed downward, and leaving the family entirely when combined upward. There is no coordinate whose weight yields $\text{shell} \cup \text{subshell}$.

Imposing arrows fragments the object monotonically. Across four cap settings and 128 edges of the Boolean lattice, **no edge decreases the component count.** And all four arrows together leave exactly one component per object — 15, 25, 19, 33, matching the object count at every setting.

Proved. A cycle under all four arrows has every coordinate non-increasing and returning to its start, so every change is zero, so $\text{tgt}(a) = \text{src}(a)$ on every step. Every cycle is a self-loop. ■

39. The arrows agree by distance in the constraint tree

The overlap of two arrows, measured against independence, ranks by their distance on $n - \ell - k - 2S$.

distance	pairs	factors at the caps of §7.4
1	$\text{occupancy} \cap \text{spin} \cdot \text{subshell} \cap \text{occupancy} \cdot \text{shell} \cap \text{subshell}$	$1.47 \cdot 1.22 \cdot 1.11$
2	$\text{subshell} \cap \text{spin} \cdot \text{shell} \cap \text{occupancy}$	$1.08 \cdot 1.07$
3	$\text{shell} \cap \text{spin}$	1.025

Strictly monotone at four cap settings, no overlap between distance classes at any of them, twenty-four measurements, correlation -0.62 to -0.75 . **The magnitudes are cap-dependent and the ordering is not** — the four-way factor runs 2.01, 1.95, 1.63, 2.62, so the factor of 2 is a measurement and the ranking is structural.

Not proved, and the general route is closed. FKG does not apply, an arrow being neither increasing nor decreasing. And positive correlation of sublattices in a distributive lattice is **not** general: 1,295 of 4,000 random pairs violate $|\mathbf{A} \cap \mathbf{B}| \cdot |\mathbf{L}| \geq |\mathbf{A}| \cdot |\mathbf{B}|$, the worst being disjoint. *What it excludes:* any proof routed through sublattices in general. A proof must use the arrow’s defining comparison.

The mechanism, offered as explanation rather than theorem. Adjacent coordinates share a bound — $2S \leq k$ on the source, $2S' \leq g$ on the target — so imposing one compresses the range available to the other. Distant coordinates communicate along the chain and the influence attenuates.

40. The precedent, and what it makes of all this

Searched by the enumerate-then-search protocol after three vocabulary collisions — *lattice*, *Cesàro* and now *monotone*, the last swamped by operator theory. In the right vocabulary the literature answers at once.

what was built here	what it is called
a weight non-increasing along every transition	a ranking function , Turing 1948, Floyd 1967
coordinates admissible, composites not	linear versus lexicographic-linear ranking functions
the maximal sub-index on which a chosen w works	a supporting invariant — restrict Q to $Q \wedge I(x)$
arrows not closing under union	disjunctive well-foundedness , Podelski–Rybalchenko
per-component analysis	standard: different SCCs take different ranking functions
min-transfer composition	max-min / subtropical , the widest-path problem

And the precedent reclassifies the object. A ranking function is a **certificate an analyst supplies**, drawn from a class the analyst chooses; the survey is explicit that every terminating relation has one and that the choice of class is what determines tractability. We fixed the rank and computed the maximal supporting restriction; the field synthesises rank and invariant together.

A certificate has nothing to predict. It is not a hypothesis about the system but a proof that the system has a property, relative to a class that was selected. **The arrows are proof technique, not physics.**

What the searches did not return is the graded measure — the overlap of two ranking functions, its ranking by distance in a constraint graph, or $E(X)$ on any of these classes. That is the same narrow residue found for 2-decomposability, for interval-dismantling and for the starshaped class. **Four times in one paper: the class is named and the count is not.**

PART X — THE PRECEDENT

41. Why the arrows agree — a decay theorem

§39 measures the agreement and does not explain it. The explanation is a theorem, and its hypothesis is the one property this framework has used throughout without naming it as a hypothesis: **the constraint graph is a tree.**

Theorem. Let X be an index whose constraint graph is a tree, carrying the uniform measure on cells, and let A_u, A_v be the arrows of coordinates u and v at tree distance d . Then

(i) the coordinates form a **Markov random field** on the tree. The indicator of X is a product of factors, one per edge — each constraint $x_i \leq \varphi(x_j)$ involves exactly one edge — so conditioning on any separating vertex gives exact conditional independence.

(ii) Hence by the **data-processing inequality**, $I(u;v)$ is non-increasing in d .

(iii) And by **Pinsker's inequality**, $|P(A_u \cap A_v) - P(A_u) \cdot P(A_v)| \leq \sqrt{(I(u;v) \ln 2) / 2}$.

Therefore **the overlap of two arrows is bounded by a quantity that decays with their distance in the constraint tree.** ■

All three parts verified.

part	test	result
(i)	conditional independence at every separating vertex	exact integer equality, 0 violations ; fails at every non-separator
(ii)	$I(u;v)$ by distance	$0.353 \cdot 0.227 \cdot 0.199$ at $d = 1$; $0.094 \cdot 0.070$ at $d = 2$; 0.020 at $d = 3$ — strictly decreasing
(iii)	Pinsker, six pairs	covariance 0.009 – 0.091 against bounds 0.083 – 0.350 , holds throughout

What is proved and what remains evidence. The **decay** is a theorem. The **strict ordering** measured at four cap settings in §39 is not implied by it: the bound is loose by a factor of three to five, and mutual information's ordering *within* a distance class differs from the overlap factor's, $\text{corr}(I, \text{factor}) = +0.62$. So the theorem establishes that agreement must fall away with distance; the measurements establish that here it does so strictly. The paper claims the first and reports the second.

41.1 The falsifier, and it fires

§4.6 requires that a test be shown able to fail. The hypothesis is treeness, so the falsifier is an index whose constraint graph carries a cycle — and the tower supplies exactly one.

stage	cells	target-side shape	$f \perp 2S' \mid g$
Λ_9	1,654	tree, $f \text{ — } g \text{ — } 2S'$	holds , 0 of 16, exact
Λ_9'	1,561	cycle , $f \text{ — } g \text{ — } 2S' \text{ — } f$	fails , 5 of 15, worst deviation 1.000

Add one bound and the Markov property is gone, the data-processing step has no chain to run along, and the Pinsker bridge has nothing to bound. **The theorem is falsifiable and falsified on demand.**

And this is the third independent result to break at Λ_9' . §29's cut law records its single violation there; §30 records it as the only stage with two two-parent coordinates; now the Markov property. §12.11.1's *the tree or the tightness* is usually read as a choice between two presentations. It is not. **It is the point at which a family of theorems stops applying**, and the 93 cells bought by closing the cycle are paid for in results rather than in cells.

41.2 What the theorem is made of

Every ingredient was already on a shelf: Markov random fields on trees, the data-processing inequality, Pinsker's inequality, and ranking functions. **What was absent was the composition.** No one had reason to ask how two ranking functions overlap, so no one assembled the three; the searches of §40 returned each piece and none of the assembly.

That is the shape of what this framework contributes, stated plainly and for the fourth time in this paper: the surrounding literatures determine **membership** — is this set 2-decomposable, is this interval dismantling, is this weight a rank, is this sequence concave — and this framework asks **how far from membership**, which requires a graded quantity that the membership question never needs. Four instances, and only in this one has assembling the pieces produced a theorem rather than another measurement.

42. The object is Brylawski's

Following §E.7's amended protocol — **search before deriving** — a precedent check was run on the concavity result. One query returned it, and it is not adjacent. It is the object.

T. Brylawski, “The lattice of integer partitions,” *Discrete Mathematics* 6(3), 201–219 (1973).

The abstract states, in these coordinates: the lattice of partitions of an integer n ordered by dominance is isomorphic to an **infimum subsemilattice**, under the component ordering, of certain **concave nondecreasing** $(n+1)$ -tuples.

Every element of Part IV is in that sentence.

derived here	Brylawski 1973
partial sums S_k of ordered masses	the dominance vector of a partition
componentwise order on $(S_1 \dots S_n)$	the component ordering
exact set = concave, non-decreasing	concave nondecreasing tuples
meet-closed, joins fail	infimum subsemilattice

Infimum subsemilattice is the correct name for what §21 proves. The one-line proof is the discrete case of a standard convexity fact — the pointwise infimum of a family of concave functions is concave — which a general search returns as textbook material.

And the join failures are now named rather than observed. Dominance on partitions of a *fixed* n is a full lattice; its join is not the componentwise maximum but the **concave envelope** of it, exactly as the stochastic-dominance literature states the corresponding fact for integrated survival functions. Componentwise max leaves the set; the correction puts it back. The measured join failures are the size of that correction.

43. What this costs, and what survives

§27.11's shape, repeated: the derivation is correct and it is not first, and a result with a fifty-year pedigree is stronger than a result proved once in a session. What it costs is the reading that any of Part IV was a discovery about three bodies. **It is a discovery that mass configurations are partitions**, and partitions have been known to be a dominance lattice since 1973. The physics supplied the question and nothing to the answer.

The mechanism is the one §E.7 already named after nine instances. *Enumerate before searching* should read **search before deriving**. Four turns of derivation preceded one query; the query took a minute. This is register 224's shape exactly, where the book's own central theorem turned out to be Baker–Pixley on the third search of a session.

What survives, stated narrowly. Not the lattice, not the semilattice property, not the concavity characterisation, not the Ravi substitution. What the search did not return is:

- $\mathcal{R}(\text{exact})$ as the tightest pairwise-monotone superset of the dominance vectors;
- its identity with the Cesàro chain $S_{\{k+1\}} \leq ((k+1)/k) \cdot S_k$, exact at nine of nine;
- the **density column** 68.1 / 38.2 / 19.8 measuring concavity against falling means;
- $E(X)$ as a graded measure on the dominance order.

That is precisely the residue §27.12 U1 already stakes out for a different object: 2-decomposability is a named class in constraint satisfaction and the *count* is not. Same gap, second object. By §27.1 the absence is not evidence, and the falsifier is one paper defining $|\mathcal{R}(X)| - |X|$ for the dominance order.

Not searched. The occupancy-clock law of Part II. It goes down as *derived here, precedent not searched*, and by §27.1 that is the only status an unrun search permits.

PART XI — THE REGISTER OF THIS SESSION

Thirty-eight entries. **A register is what makes a theory checkable rather than a claim**, and it is printed here for that reason: every result above is stated with the conditions under which it holds, and every statement that failed those conditions is recorded below with what it excludes. §2.15's rule — a failed attempt is a bound — is why this section is evidence rather than apology.

1. Bracket edges labelled by δ order. In the Sc VI recomputation the interval endpoints were labelled lo/hi by increasing δ , giving a negative width. E is *decreasing* in δ . Values were exact throughout; the labels were reversed and the sign caught it. Excludes nothing about the book.

2. Deletion and insertion offered as paradox handling. §16.8's attack table — delete → repaired, insert → amplifies — was presented as the book's treatment of the grandfather paradox. Those results are tamper detection. The book's treatment of paradox is §30.4.1 and the absence of negation, which was stated third and buried. The framing was imported to make an existing table look like an answer.

3. A §4.6 probe whose verdict was written before its number. A falsification probe for the decomposability test printed "*the test returns <100%, so it can fail*" in the same string as the computation. It returned 100%; the probe demonstrated nothing. A correct probe fails — 0/1, 0/5, 1/50, 134/300 on random subsets. **§26.1 committed inside the check written to prevent it**, which is register 310's shape.

4. A.18 applied to a region carrying a congruence. Predicted system B would fail meets only. It fails both. §12.11.2 already states that imposing the parity congruence kills the join closure. Converted to a bound at §18: the join-closed half cannot reach any integer-valued index.

5. "Coincidence" asserted in a closed index. The $204 = 204$ crossing was ruled a coincidence after a bijection was refuted. Two monotone integer sequences of like degree cross exactly once; the crossing is forced and its location determined. The refutation was correct and the conclusion drawn from it was not.

6. A loose envelope taken for the tight one. System A used $M \leq 2S$ where $m_3 \leq m_2 \leq S/2$ gives $M \leq (3/2)S$, single-parent and containing. Every density in §17 understates by about a third. Same class as Λ_9' — an admissible tightening left untaken because the loose bound closed.

7. $\mathcal{R}$ predicted to beat the hand-built chain. Predicted $\mathcal{R}(\text{exact})$ strictly smaller at $n \geq 4$ on §D.4.3's precedent. It is equal, nine of nine. The path of adjacent Cesàro bounds is already transitively closed, so §D.4.3's phenomenon — a recovered system strictly richer than the defining one — does not occur here. The failed prediction is what established that the construction was tightest.

8. Four turns of derivation before one search. §E.7's protocol reversed. The whole of Part IV was derived and then found in Brylawski 1973. Cost: a session. The correction is the protocol, already written.

9. Six references listed and never cited. Birkhoff, Dilworth, Nesterov and Nemirovskii, Racah, Rival and Siggers were carried into the reference list from *The Method's* bibliography into a paper that does not use them. Audit 7 exists for exactly this, and it is the failure the book calls **dishonest** rather than wrong — a list implying use. Birkhoff is now cited where the down-set correspondence is used; the other five are removed.

10. Reachability figures stated without their caps. §8's 67.9%, 24.7%, 11.4% and 0.4% are computed at the caps of §7.4 and the section did not say so, while §7's own table states four cap settings a page earlier. Register 292's shape — a figure printed without the coordinate list it depends on. Corrected in place.

And two flags that were detectors and not defects, localised before recording per §4.7: audit 15 counted table rows across the whole document rather than within Appendix B.1, and audit 20 compared a straight apostrophe in the source against pandoc's typographic one in the artefact. Four of the last six flags this paper has seen were the detectors and not the paper, which is the ratio §3.9 reports for the book.

11. Five of the twenty audits cannot be run, because they are not named. Running the set against this paper required first recovering it from *The Method* 1.3, which never lists the twenty in one place. §3 names seven; §3.8 adds DEPTH, SEQUENCE and PROJECTION; the registers add ARITHMETIC (14), ENUMERATION (15), FIDELITY, SCOPE and ANTECEDENT. **Fifteen in all.** Audits 8–13 and 16–18 exist only as a count. They are recorded here as unrunnable rather than as passing, on §E.4.1's own ruling that an assertion is not an enumeration — the ruling the book applied to Q and did not apply to its audits.

12. The occupancy clock claimed as a law before it was searched. Part II's theorem was derived, tested at four cap settings, written up, and only then searched. The content is standard — DAG gradings, semilattice-graded semigroups, and *semigroupoid* as the name for Λ_9 's type outside its top era. §E.7's protocol reversed for the second time in one session, and the tenth entry in §E.5's ledger. §6.1 records the demotion; the narrow residue is the

derivation from $g \leq q \leq k$ and not the graded object.

13. The balance predicted to hold at one cut and holding at four. Predicted that $\sum_v |A(v)| \cdot |B(v)| = |\Lambda|$ only at the transfer. It holds at ℓ , q , f and g — every two-sided cut of the tree. The factorisation is a property of the **tree**, not of the transfer, and §12.6.1 attributes it to the transfer. A prediction refuted into a sharper reading.

14. Opposition attributed to two parents. Stated that opposed monotonicity fails at two-parent coordinates. True at Λ_8 and false at every stage above it. The cause is that at Λ_8 the coordinate g has two parents **and no child**; from Λ_9 onward it has the child $2S'$ and opposition returns. The right exception with the wrong reason, and the third prediction refuted in this session.

15. A criterion stated with irreducible where the proof needs prime. The step law's first form disagreed with the truth on 35 of 116,138 intervals — exactly the intervals with $\perp$ or $\top$ as an endpoint. The property the proof uses is primeness, which the extremes satisfy trivially and which convention excludes them from. Corrected to 0 disagreements.

16. A cut test that skipped the case it was built for. The first run of the cut law tested only two-sided cuts, and g becomes a three-way cut from Λ_9 onward — so the coordinate the test existed to examine was silently never examined. §4.6's shape: a check that cannot see a case returns no failures on it.

17. A crossing called a coincidence. The $204 = 204$ agreement between Λ_8 's exact set and B 's triangle gap at $N = 16$ was ruled coincidental after a bijection was refuted. Two monotone integer sequences of like degree cross exactly once; the crossing is forced and its location determined. The refutation was right and the conclusion drawn from it was not.

18. Six more detector literals, in one rebuild. Expanding the paper broke five checks and none of them was the paper: a single-digit regex for figure references and the same literal again for captions, so figures 10–12 were invisible; a hardcoded image count of sixteen; a part name the renumbering moved, which truncated the body and made two references look uncited; and an appendix boundary. Two real defects surfaced beside them — a broken O-item enumeration and fifteen cap-bearing figures without their caps, both repaired. **The instruments have now needed eleven repairs against the paper's five.** Registers 305, 310 and 337 record the identical mechanism three times in *The Method 1.3*; this session has produced eleven more, and the rule is worth stating flatly: **an instrument written against an object will encode the object's current size, and the next edit is what finds it.**

19. A figure caption absent from the artefact, and no audit could see it. Figure 3's caption — label and body — did not render at all, while the source carried it and every source-side check passed. The cause was float migration: pandoc's implicit-figure conversion makes each image a LaTeX float, two landed on one page, and the second caption was lost. **COHERENCE compared figure references to captions in the source, so it verified that twelve captions existed in the file where they had always existed.** Register 190 is the same failure — the operator this book is named for printed as 107 blank boxes across 44 pages while every audit reported clean, because all of them read the source and none read the page. Repaired by building the images inline rather than as floats, and the audit now reads the artefact for every caption label and body.

20. A closed form reported from a broken sampler, and caught by a sanity check. $O3$'s limit was computed as a cone-volume ratio and came out at $1/n!$ across five values of n — 49.93, 16.68, 4.14, 0.83, 0.14 per cent. The sampler made the differences exchangeable by construction, so $1/n!$ measured the sampler. The lattice density is 61% where the “limit” said 17%, and comparing the two is what caught it. **A closed form that agrees with itself across five points is not thereby a result.**

21. Adjunction tested against a set that already failed. $O8$'s first run adjoined coordinates to the *exact* n -body set, which carries 224 join failures of its own, so every candidate registered as inadmissible including two coordinates the object already has. $E2$ is a statement about closed sets and the baseline must be one. Re-run against the envelope, the answer separates cleanly and confirms the committed prediction.

22. A thirteenth detector literal, in the check that counts the register. CONSISTENCY located the register by the string *PART VI*, and the part renumbering moved it — so the audit that verifies the register's own count could no longer find the register. Re-anchored on the heading text rather than on a numeral. **The instruments have now needed thirteen repairs against the paper's seven**, and every one has been a literal or a wrong artefact. That ratio is the most reproducible finding in this paper.

23. Four searches run against a whole statement, where the protocol says enumerate first. $O6$'s precedent search was run four times as a single query on the whole occupancy-clock claim, returning adjacency each time. Decomposing the claim into six target cells and searching per cell — §2.10's *enumerate targets from the index before searching* — located a fourth immediately: min-transfer composition is **max-min composition in the subtropical semiring**, and the induced path quantity is the **bottleneck**, or widest path. **The failure was not in the searching**

but in never having enumerated, which is §28.4's defect committed by the searcher and the reason §2.10 gained its second clause. The residue falls from an undifferentiated gap to two named cells.

24. The step law derived and found in print, and a sampled closure standing for four turns. §27's criterion is **interval-dismantling**, named, with the singleton case being exactly the doubly irreducible elements — §28 restated. Rival's theorem is the shape. The literature's relativised primeness and this paper's absolute form agree on Λ_8 at 0 disagreements each over 116,138 intervals, so the paper's is a specialisation and the general form is the one to cite. Eleventh in §E.5's ledger, and located by the same enumerate-then-search protocol that resolved O6 — **two precedents in two runs, after four whole-statement queries had returned adjacency**. In the same pass the source-seniority Λ_{10} of O1 was verified exhaustively at 15,260,050 pairs, having stood on a 1,200-cell sample: §2.7 requires the exhaustive test where it is affordable and it was affordable throughout.

25. A second doubly-occupied vocabulary, and the class was named in 1962. O7's envelope — sequences whose Cesàro means fall — is the classical **starshaped** class, and *convex* $\subset$ *starshaped* $\subset$ *superadditive* is a strict inclusion proved by Bruckner and Ostrow in 1962, with a discrete literature since. The first search returned only summation theory: **Cesàro is doubly occupied exactly as §27.10 reports for lattice**, and the query had to be reformulated in the right vocabulary before the shelf was reachable. §20's density column is the measure of a strict inclusion that has been known for sixty-four years. Twelfth in §E.5's ledger, and the third precedent located in three runs of §2.10's enumerate-then-search protocol.

26. The dual taken as superadditive where it is subadditive. Testing whether the third class of the Bruckner–Ostrow chain is admissible, I imposed superadditivity on concave objects. The dual of concave is **subadditive**, the chain came out broken, and the classification would have been reported wrong had the inclusion not been checked. The closure verdict survived — sums break closure in either direction — but the class did not. *What it excludes*: reading a dualised hierarchy off its primal statement without dualising every member.

27. A poset asked a linear question, twice. The five classes around the envelope were tested as a chain and reported as broken at two different operators, each time producing a plausible-looking “wrong operator” diagnosis. Measuring the full containment matrix instead shows **every link holds and two classes are simply incomparable** — *A(f) concave* and *f/x falling*. The object was a poset and the test was linear. Register 168's shape: a conclusion drawn from a structure the test could not represent, and the remedy was to measure containment rather than assume order.

28. The density limit derived after being bounded, and by a theorem rather than a trend. O5 stood at *the limit exists and lies in [0, 60.298%]*. With O7's identification the object became a ratio of lattice points in two rational cones, Ehrhart's theorem applies, and the limits are exact: **1 at n = 2, 3/5 at n = 3, 48/185 at n = 4**. The bracket held and the value arrived inside it. **The earlier cone-volume attempt needed no sampler at all** — the failure recorded at entry 20 was not a hard problem badly approached but an easy one approached with the wrong instrument.

29. One open item closed another. O5 was not closed by more measurement. It was closed by O7 naming the envelope, after which the right theorem was obvious. **Two items in the same appendix, one blocking the other, and neither entry said so** — which is §2.16's grid failing to be read on the paper's own list of ignorance.

30. A repair stated in prose is not a repair. O4 stood for several revisions as *repairs stated, not applied*, which is §4.3's failure — a claim of completion that had not happened — dressed as an open item. It is now four substitutions anchored on strings occurring once each, applied to a copy and checked, with the invariant stated first. **And building it found something prose had hidden**: two of the four figures also appear inside register entries 296 and 337, where they are correct as history, so a find-and-replace repairing the body would have corrupted the record. The distinction only became visible when the repair was made executable.

31. A fourteenth detector literal — the number-word list stopped at twenty. CONSISTENCY spells the register's stated count to compare it with the entries present, and the spelling table ran one to nineteen with hyphenated compounds. At thirty entries it returned None and the audit failed on a correct paper. Generalised to ninety-nine. **The instruments have needed fourteen repairs against the paper's eight**, and this one is the purest instance of the mechanism: a table sized to the object as it stood when the table was written.

32. The tight-K discrepancy exhausted rather than solved. Eighty candidate quantities across four cap settings, two stages and six projections; none equals 2,475 and the nearest is off by 133. **What the sweep establishes is not the answer but the shape of the remaining question** — no quantity the construction produces matches the figure, so the phrase denotes something outside what the book prints, or the figure is wrong. §2.15: the failure is a bound, and the bound is that the next attempt need not search the computable space again.

33. O9 given a candidate, and the candidate given its own disqualifier. The composition law predicts that no cascade raises occupancy — a statement about *chains* through the index, which $E(X) = 0$ does not govern, unlike

statements about its contents. That is testable against public line lists. **And it is worthless if $g \leq q \leq k$ is definitionally true of how cascades are recorded**, in which case the test cannot fail and §4.6 disqualifies it before it runs. Establishing which is the first work, and it is not done here — so this is recorded as a candidate, not as a test.

34. Register entries written for edits that had not been applied. The pass closing O3, O9 and O10 raised on its last substitution and wrote nothing, while a second pass added the register entries describing all three. **For one build the register described work the paper did not contain** — §4.3 exactly, a claim of completion verified by nothing, and the failure mode the audits cannot see because CONSISTENCY compares counts and not content. Caught by probing the paper for the strings the entries claimed. **A multi-part edit must be written atomically or verified per part**, and this one was neither.

35. The arrow was an assumption, and the candidate that died found it. O9's cascade test was disqualified because $g \leq q$ reads g as *placed* while $g \leq 2(2f+1)$ reads it as *present*, and the two agree only when the destination begins empty. Building the index that distinguishes them — §17.4's repair, (g, G) with $g \leq G \leq 2(2f+1)$ — gives a closed object at **13,775 cells and 94,868,425 pairs with zero failures**, in which, at the caps of §7.4, **31.6% of composable steps raise occupancy**. Λ_0 's clock is a property of the empty-destination assumption, not of the object. §7.1 states seven constraints, four origins and *nothing else*; this is the second thing it needed, after register 301's $k \geq 1$, and it is a conflation rather than an omission. Part II's §11 carries it and the abstract states it.

36. A limitation found by refusing a weak candidate rather than by planning. The disqualifier was flagged when the candidate was written and left as *not established*. Establishing it took one step, killed the candidate, and produced the strongest result in the paper about its own object. **§4.7's discipline run forward**: a flag is a hypothesis, and localising it is not optional even when the flag is one's own.

37. O9 closed in both halves, and the second half dissolves it. The reframed question — does any temporal structure survive occupied destinations — is negative on the whole object, which is **one strongly connected component**, and positive on the **maximal non-accretive sub-index**, which is closed and three times Λ_0 . Empty destinations are sufficient and not necessary. And the precedent settles the original half: the arrow is a **ranking function**, the sub-index a **supporting invariant**, and **a certificate has nothing to predict**. Recorded because the item changed meaning three times and only the third framing was answerable.

38. Fifteen turns of work absent from the paper. The arrow family — one per coordinate, the trichotomy, the meet-semilattice, the tree-distance correlation, the fragmentation results and their precedents — was derived, swept, proved and searched without any of it reaching the document. **A result not written down is a result the audits cannot see**, and CONSISTENCY, COHERENCE and PROJECTION all passed throughout on a paper missing its largest part. Now Part IX. The mechanism is §4.3's: describing and doing feel identical while working.

What the register shows. Nineteen of thirty-eight are the same mechanism in different clothes: a statement written before the number beneath it was read, or a prior result applied past the boundary it states. Two — 4 and 7 — are failed predictions that produced results, which is §2.15 operating as intended. **None was caught by a reader.**

APPENDICES

Appendix A — Computational methods

Python 3, no external dependencies beyond the standard library. All computations are exhaustive over the stated sets unless a sample size is given; no sampling was used in any reported figure.

$\mathcal{R}$, implemented from §6.1. Value sets $\hat{A}_i(X) = \{x_i : x \in X\}$; envelopes $\hat{\phi}_{ij}(v) = \max\{x_i : x \in X, x_j \leq v\}$; then $\mathcal{R}(X) = \{y \in \prod \hat{A}_i : y_i \leq \hat{\phi}_{ij}(y_j) \text{ for all } i \neq j\}$. For $d \geq 4$ the product is enumerated with prefix pruning on all pairs already fixed.

Λ and the tower, built by nested loops over the constraints of §7.1 with the caps of §7.4, the eighth condition $k \geq 1$ of register 301, and the coupling bounds $2S' \leq g$, $2S' \leq 2f+1$, $2S' \leq v \leq g$, $2J_c \leq \hat{\phi}(k)$ with $\hat{\phi} = \{1:3, 2:4, 3:5\}$, $2K \leq 2J_c + 2f_{\max}$, $|2J - 2K| \leq 1$.

Composition, per §12.11.0: $b \rightarrow a$ defined when a 's target equals b 's source, the composite running a 's source to b 's target and transferring $\min(q_a, q_b)$ — **bottleneck composition in the subtropical semiring**, per Appendix C O6.

Strongly connected components by Tarjan's algorithm on the object graph, objects being the 4-tuples $(n, \ell, k, 2S)$ and $(e, f, g, 2S')$ identified as shapes.

Single-deletion repair screened first for cells that could alter a value set or an envelope maximum — a cell can change $\mathcal{R}$ only if it uniquely holds a coordinate value or uniquely attains some $\hat{\phi}_{ij}$ — with full recomputation for any survivor. Zero survivors on Λ_8 .

The n-body index: exact set enumerated as partial-sum vectors of non-increasing non-negative integer sequences with total $\leq N$; envelope by the Cesàro chain.

Computation is cut into pieces, and this is a protocol rather than a habit. Every computation in this paper is executed in a **Zeno structure**: the smallest instance first, one cap setting per invocation, one stage of the tower at a time, and closure generation run incrementally against a frontier rather than over the whole set each round. Each piece is verified before the next is written.

The rule exists because it was measured rather than preferred. Four computations in this work were first attempted whole and exhausted their time budget — the sublattice generation of §38, the normalised-amplification sweep, the interior-operator verification, and the first closure test of §11's 13,775-cell index. Every one succeeded on the first attempt once re-cut smaller. **A run that fails whole costs the whole run; a run that fails in pieces costs one piece.**

Two corollaries have both paid here. **Check whether the piece is already in hand** — §38's fragmentation monotonicity at the book's caps was read off a table already computed, at no cost. And **check whether it is provable** — the trichotomy of §37 and the fragmentation theorem of §38 each took one line and between them deleted four planned cap sweeps. Proving before measuring is the cheapest piece of all.

Appendix B — Every figure in this paper, indexed

quantity	value	where
tower cell counts	$976 \cdot 1,654 \cdot 1,561 \cdot 2,535 \cdot 13,585 \cdot 70,905 \cdot 199,130$	§2
ambient boxes	$6,912 \cdot 27,648 \cdot 27,648 \cdot 110,592 \cdot 663,552 \cdot 5,308,416 \cdot 47,775,744$	§2
fill	$14.12\% \rightarrow 0.42\%$	§2
$E(X)$, four stages	0	§3
refused, Λ_8	5,936	§3
pairs tested on Λ_8	475,800, zero failures on three tests	§3
$F(1)$, $F(-1)$	976, 2	§3
rank levels, widest, mean	18, 122 at rank 11, 11.0666	§3
generators, covers, $\Sigma(A_i -1)$	17, 20, 17	§3
reflections	$8 \cdot 112$ with 0 fixed	§3
cylinder sections	$165 \cdot 330 \cdot 345 \cdot 136 = 976$, $\langle q \rangle = 1.4631$	§3
Sc VI	736,688 in [735,860, 737,380]	§3
register share of the book	20,074 of 94,354 words = 21.3%	§4
composable pairs	41,682, zero failures	§7
occupancy rises	0 at four caps	§7
eras	$8 \cdot 15 \cdot 10$, pure in k	§7
identities · reverses · inverses	$8 \cdot 389 \cdot 64$	§7
conservative cells	904 of 1,654 = 54.7%	§7
object reachability	739 of 1,089 = 67.9%	§8
cell-pair reachability	675,606 of 2,735,716 = 24.7%	§8
lattice comparability	$311,180 = 11.4\%$; both = 9,720	§8
deletion repair	976 of 976, 0 critical cells	§9
decomposable cells	1,654 of 1,654 = 100%	§9
second bridges	20 tested, 0 closure failures, defects 23–354	§13
cut vertices	5, one separating	§12
three-body A density	$59.4 \rightarrow 51.4\%$ loose, $77.4 \rightarrow 68.1\%$ tight	§17, §20
three-body B density	$32.8 \rightarrow 26.9\%$	§17
B gap decomposition	triangle + congruence + cap, exact at four caps	§17
pure triangle	0 join failures at four caps	§18
triangle + parity	26,328 join failures at cap 20	§18
P1	$165 \cdot 455 \cdot 969 \cdot 1,771$, $E = 0$, density 100%	§19
n-body density	$68.1 \cdot 38.2 \cdot 19.8\%$ at $N = 20$	§20
$\mathcal{R}(\text{exact}) = \text{chain}$	9 of 9, as sets	§22
synthetic clock test	0 rises / 110 rises	§6

Appendix B.1 — Figures

Thirteen figures, all generated at **300 dpi** from the computations of this paper by the code of Appendix A. Register 194 of *The Method 1.3* records fifteen inherited figures at a median of 164 dpi, below print standard, and states that the audit’s value was writing the number down rather than repairing them. This paper’s figures are at standard, and the class is closed for the new material.

figure	subject	section
1	the tower, fill and closure	§2
2	Λ_9 ’s eras and the occupancy clock	§7
3	the two orders, and their overlap	§8
4	the hypothesis, and what the repair costs	§11
5	every second bridge, priced	§13
6	the two presentations’ failure sides	§17
7	the envelope gap, decomposed	§17
8	density in n , and the tightening	§20
9	concavity against falling means	§21
10	the clock’s tick, decomposed	§25
11	pushback by rank	§26
12	the step, and double irreducibility	§27
13	the two indices and their agreement	§35

Appendix B.2 — The audit run

The audit set of *The Method 1.3* §3, recovered and run against this paper’s source and its rendered artefact together. Reproducible: `audits.py`, which exits with the number of failures.

#	audit	verdict	evidence
1	LATTICE	pass	475,800 pairs, 0 join/meet and 0 modularity failures; $F(1) = 976$, $F(-1) = 2$; $\Sigma(A_i - 1) = 17$; $\Sigma A_i \cdot B = 976$
2	EQUATIONS	pass	9 of 9 equations evaluated true against the construction
3	CONSISTENCY	pass	the register’s stated count and its entries agree
4	REDUNDANCY	pass	0 statements surviving their own restatement across 41 sections
5	ARTEFACT	pass	20pp, 0 bad glyphs, 0 unembedded fonts, 16 images, 0 overflowing pages
6	COHERENCE	pass , after repair	every caption label and body checked in the artefact , not the source; register entry 19
7	ATTRIBUTION	pass , after repair	7 works listed, 7 cited; register entry 9
14	ARITHMETIC	pass	Λ_8 , Λ_9 , Λ_{10} rebuilt from §7.1; shown able to fail
15	ENUMERATION	pass , after a detector repair	every stated count matches what it counts
19	SEQUENCE	pass	§1–§24, no breaks in ascent, no duplicates
20	PROJECTION	pass , after a detector repair	40 of 40 headings project into the artefact
21	DEPTH	pass	max depth 3, 0 skipped levels
22	SCOPE	pass , after repair	57 percentage claims, 0 cap-bearing without a stated cap; register entry 10
23	FIDELITY	pass	10 source strings present in the artefact’s text layer
24	ANTECEDENT	pass	0 numbered sections without a parent part
8–13, 16–18	<i>unnamed</i>	unrunnable	not named anywhere in <i>The Method 1.3</i> ; register entry 11

And the set was run again after this appendix was written, because writing it changed the object. Two audits failed on that pass and both were instruments carrying literals — a hardcoded list of number words that had no entry for *eleven*, and a section boundary that stopped at Appendix C and so swallowed this table. **Five instrument repairs against three paper repairs**, and every instrument fault was the same one: a literal narrowing a check to the cases foreseen when it was written. Registers 305, 310 and 337 of *The Method 1.3* record the identical mechanism three times, which is the strongest evidence in this paper that the mechanism is real and not anecdote.

Twenty run, twenty pass, none unrunnable. *The Method 1.3* gives twenty as a count and names fifteen — §3 names seven, §3.8 adds DEPTH, SEQUENCE and PROJECTION, and the registers add ARITHMETIC, ENUMERATION, FIDELITY, SCOPE and ANTECEDENT. **The remaining five were unnamed and are defined here**, each against a defect class this session actually produced rather than invented for the count:

#	audit	what it catches	motivating register entry
8	CLAIM	a register entry naming a section or open item that does not exist	34, 38
9	PRESENCE	text present in the source and absent from the rendered page	19
10	INSTRUMENT	a numeric literal in the audit script equal to a count of the document's own structure	18, 22, 31
11	HYPOTHESIS	a universal-form result with no hypothesis stated, or a hypothesis with no fired falsifier	—
12	PROMOTION	a pattern called a theorem without a proof, or a proof not labelled as such	13, 14, 20, 26, 27

Defining them is §E.4.1 applied to itself: an assertion is not an enumeration, and the remedy for a count without names is names. Three of the five failed on their first run and all three failures were in the new instruments — PRESENCE's normaliser left a space before punctuation after stripping markup, INSTRUMENT flagged mathematical constants that are verified by recomputation rather than hardcoded, and HYPOTHESIS read an explicit *no hypothesis needed* as a missing one. **That is the fifteenth, sixteenth and seventeenth instrument repair against the paper's eight.** PRESENCE then needed four more before it passed — a space left where a combining diacritic was stripped, a probe pool and an artefact stripped with different character classes, and two rounds of whitespace normalisation. **Twenty-one instrument repairs against the paper's nine.** The audit built to catch text that never reaches the page could not itself compare two renderings of the same sentence, and every one of its four faults was an asymmetry between how it treated the source and how it treated the artefact — which is, exactly, the defect class it exists to detect.

Appendix C — Open, closed, and unclosable

Every item is classified on one ordered coordinate, and the classification is exhaustive.

fully closable < closable to a limit, then an observation < unclosable

The third value had to be offered, not found. Indexed on (closure kind, obstacle, cost) the printed set gives **E = 3** and every admitted-and-absent cell lies in the first two kinds. It cannot lie in the third, because no item carried that value and $\mathcal{R}(X) \subseteq \prod_i A_i(X)$: **the operator names cells and never a value the index lacks**, which is §30.1.4 demonstrated on this paper's own list of ignorance. Offering the third kind grows the ambient box from 12 cells to 27 and leaves E at 3 — the same three gaps in a box more than twice the size, which by register 275 makes the shortfall a stronger statement, not a weaker one.

Three cells the structure admits and this appendix does not contain: *fully closable · retrievable · days*; *closable to a limit · buildable · days*; and *closable to a limit · buildable · unbounded*. The last pair is the informative one — every limited item here is a **search**, and none is a **construction that runs out**. §12.11.0.2's bounded fourteenth axis is exactly such an object and belongs here.

Fully closable — and three are now closed

O1 — composition above the ninth axis. CLOSED, and the answer overturns the instance rather than the law. §12.11.0 records composition lost at Λ_{10} because seniority v has no source counterpart. Supplying one — a source seniority w with $2S \leq w \leq k$, by the identical construction that gave the ninth axis $2S' \leq g$ — gives a 5,525-cell

index that is closed, whose targets are legal sources 65 of 70, and which **composes on 266,125 pairs with zero failures**. Closure is now exhaustive rather than sampled: **all 15,260,050 pairs tested, 0 join and 0 meet failures**, and independently $\mathcal{R}$ swept over the 1,048,576-cell ambient box returns exactly 5,525, so $E(X) = 0$ by two disjoint routes. So the loss of composability at the tenth axis is **a defect of the construction, not a law**. Register 314's *an axis can be exact and not composable* survives; its instance does not.

O2 — an observable adjoined under E2. CLOSED, and one prediction failed first. Tested against the closed envelope of the n-body index:

adjoined	join fail	meet fail	admissible
an existing coordinate — m_1, S, M	0	0	yes
$\max(m_1, S) \cdot \min(S, M) \cdot a$ constant	0	0	yes
$S_{13} = M - S + m_1$	1,935	1,935	no
$S_{23} = M - m_1$	4,350	4,350	no
$m_2 = S - m_1$	4,171	4,171	no
$\min(m_1, M - S)$	3,638	3,638	no

Of the three pairwise sums, only the one §17.4's repair made a coordinate is adjoinable. The other two are *differences of the coordinates*, and A.6 forbids them — which is why §16's filter table can separate three masses exactly with three pairwise sums and no single closed index can carry them. The last row is the nuance: $\min$ is a meet-morphism and still fails, because its argument $M - S$ is a difference. **The forbidden form propagates through the morphism.**

O3 — the tight-K discrepancy. Exhausted computationally; the remaining step is not a computation. Four routes give 15,150 at Λ_{12} and 45,450 at Λ_{13} against §12.11.5's 2,475. A systematic sweep then computed **eighty candidate quantities** — cells cut, factorisation defect, per-section defect at each transfer, distinct (J_c, K), (f, J_c, K) and A-side projections of the cut, and cells sitting below the cap — **across four cap settings and both stages. No quantity equals 2,475**; the nearest is 2,608, off by 133. What does equal it is 3.5% of Λ_{12} 's cell count, to within seven cells.

So 3.5% of *the product* denotes something this rebuild cannot reconstruct from the printed definitions, or the figure is wrong. **Settling that requires what was intended, which is the author's and not a computation's.** Recorded with the sweep so the next attempt need not repeat it.

O4 — the three defects of §4. CLOSED as a verified patch. The repairs are no longer stated in prose; they are four exact substitutions, each anchored on a string occurring **once** in *The Method 1.3*, applied to a copy and checked, per §2.19's safe-repair standard — invariant written first, invariant computable, edit a substitution and not a rewrite, reversible, checked afterwards.

	section	line	why
P1	§30.1.3	7167	<i>Part I sits behind the mathematics it governs</i> — register 321 moved it
P2	§4	1047	<i>three hundred and twenty-eight</i> → three hundred and forty-two
P3	§4	1050	<i>fourteen entries collapse to seven mechanisms</i> → twenty-nine , per registers 309 and 310
P4	F.4.1	8899	<i>Chapters 1 to 25 without it, 19,504 : 38,817 = 0.502</i> → 20,074 : 74,280 = 0.270 , the range having been redefined by register 341

And two places that must NOT be changed. The same two figures appear inside register entries 296 and 337, where they are **history and correct as history** — §26's own distinction, that a procedure states what to do and the record states what happened. A naive find-and-replace would corrupt the register while repairing the body, which is why the patch is anchored on body context and verified to match once each.

Verified after application: the four stale strings are gone, the two register occurrences survive, and the word count moves by ten. Closing it is now one pass by the author, which is what *fully closable* was always supposed to mean.

Closable to a limit, then an observation

O5 — the density trend. CLOSED, to values. O7's identification made the right instrument obvious: the density is a ratio of lattice-point counts in two **named rational cones**, so Ehrhart's theorem applies. The counts are quasi-polynomials of degree n whose leading coefficients are the cone volumes, and the limiting density is their ratio — **interpolation at known degree, which is a theorem, rather than extrapolation from a trend, which §20.1.2 forbids.**

n	vol(concave)	vol(starshaped)	quasi-period	limiting density
2	1/4	1/4	2	1 — the classes coincide
3	1/36	5/108	6	3/5 = 60%
4	1/576	185/27648	12	48/185 $\approx$ 25.946%

And it lands inside the bracket this paper already carried. O5 previously stated the $n = 3$ limit as existing and lying in $[0, 60.298\%]$, measured at $N = 600$. The derived value is exactly **3/5**. That is §23.6's own shape — a bracket later occupied by a deduction, the interval holding and the value arriving inside it.

The sequence 1, 3/5, 48/185 is not extended here. Its general term would come from the pattern the values make, and §20.1.2 forbids that; $n = 5$ is computable by the same route at a larger period and a heavier count.

What closed it was O7, not more measurement. Until the envelope was identified as the classical *starshaped* class, the density was a trend in a construction of ours and the one attempt at its limit produced $1/n!$ from a broken sampler. Once it was a ratio of two named cones the instrument was obvious and no sampling was needed at all. §2.17 in the plainest form: locating one thing gave the fourth.

O6 — the occupancy law's precedent. FOUR OF SIX CELLS LOCATED, by enumerating the target set before searching rather than querying the whole statement, per §2.10 as amended.

cell	the component	status
1	a weight decreasing along every edge forbids cycles	located — an $\mathbb{R}$ -grading on a DAG
2	a semigroup graded by a semilattice, $S_a \cdot S_b \subseteq S_{\{a \wedge b\}}$	located
3	a category lacking identities	located — <i>semigroupoid</i>
4	composition transferring min of two weights	located — max-min composition; the subtropical semiring ($\mathbb{R}, \min, \max, \infty, -\infty$); the widest-path problem
5	cycles confined to level sets, identities only at the maximum	not located
6	counting constraints <i>supplying</i> the grading	not located

Cell 4 is the substantive find and four whole-statement queries had missed it. Λ_9 composes by transferring $\min(q_a, q_b)$, which is exactly max-min composition, and the path quantity it induces — the least weight along a path — is the bottleneck, or widest-path, problem. **So q is a capacity and a composite carries the narrowest one.** §12.6.1 calls q *the conserved coupling*; the literature's reading is sharper and this paper adopts it. §7's 41,682 composable pairs are a transitive closure in the subtropical semiring, and register 313's *composition is time* separates into two named objects — a grading, which is the clock, and a bottleneck composition, which is the transfer.

And the method is Chapter 19's, applied to itself. Four earlier searches ran a whole statement against a search engine and returned adjacency. One run of §2.10's protocol — enumerate the target cells from the index, then search per cell — returned a literature. That is the retrieval chapter's own claim demonstrated on the paper's own ignorance, and it is why the residue is now two cells rather than an undifferentiated gap.

O7 — E(X) on the dominance order. CLOSED as far as a search can close it: four of five cells located, and the structural question answered.

cell	the component	status
1	dominance on partitions is a lattice	located — Brylawski 1973
2	its representation as concave nondecreasing tuples	located — Brylawski 1973
3	the envelope, S_k/k non-increasing	located — the <i>starshaped</i> class

cell	the component	status
4	$E(X) = 0$ as a class	located — 2-decomposability, §27.12 U1
5	$E(X)$ as a graded measure	not located

Cell 3. The condition S_k/k non-increasing is the classical **starshaped** condition, defined exactly that way, and **Bruckner and Ostrow (1962) proved the inclusions $\text{convex} \subset \text{starshaped} \subset \text{superadditive strict}$** , with a discrete literature since as the *hierarchy of convexity of sequences*. Dualised for concave objects the chain is $\text{concave} \subset \text{starshaped} \subset \text{subadditive}$. So §22's exact set and its envelope are two members of a classical hierarchy, and §20's density is **the measure of a strictness known since 1962**.

And the structural question the identification raised is answered: of five classes in that neighbourhood, exactly one is a closed index, and it is the one this paper uses. Measured at $n = 4$ $N = 12$, $n = 5$ $N = 9$ and $n = 4$ $N = 16$, stable:

class	cells	join fail	meet fail	$E(X)$	admissible
concave — the exact set	155	1,073	0	186	no
$A(f)$ concave — <i>concave in the mean</i>	532	11,874	0	440	no
f/x falling — starshaped	341	0	0	0	yes
subadditive	484	0	487	31	no
$A(f)/x$ falling	723	0	19,538	104	no

The failure sides sort by form, and the sorting is §12.11.2's. The two concavity classes are **difference** bounds and fail joins only; subadditive and $A(f)/x$ falling are **sum** bounds and fail meets only; starshaped is a **ratio in one variable** and fails neither. The book's excluded-forms table and a sixty-four-year-old function-class hierarchy agree exactly, and neither literature knows of the other.

The starshaped envelope is not one choice among several. It is the unique admissible member of its own classical neighbourhood, and the density measures the distance from the exact class to it.

The five classes are a poset, not a chain. Every link needed holds — concave is contained in all four others, and f/x falling $\subseteq$ subadditive $\subseteq A(f)/x$ falling — but $A(f)$ concave and f/x falling are **incomparable**. Bruckner–Ostrow's linear presentation is of one branch.

Cell 5 is what remains. The class is named, the inclusions are known strict, and no source returned by these searches **counts the gap**. Searched under *convex versus starshaped enumeration, proportion, strict inclusion measure*; the results are adjacent — convex sequences and combinatorial counting, Chebyshev inequalities for star-shaped sequences — and none is the quantity. §27.1: the absence is not evidence. *Falsifier*: one source giving the proportion of concave sequences among starshaped ones, at any n , or defining $|\mathcal{R}(X)| - |X|$ for the dominance order.

And the second strictness is a quantity this paper did not have. At $n = 4$, $N = 12$: concave/starshaped **45.5%**, starshaped/subadditive **70.5%**, concave/subadditive **32.0%**. The first is the price of the envelope; the second is the price of the boundary the language cannot cross.

O8 — the step law's precedent. LOCATED, by the same enumerate-then-search protocol as O6. The criterion is in print with a name: an interval $[u, v]$ is called **dismantling** when v is infimum-prime in the filter of u and u is supremum-prime in the ideal of v , with u, v not the extremes — and the literature notes that dismantling intervals of a single element are **precisely the doubly irreducible elements**, which is §28 of this paper. The shape is **Rival's**: every sublattice of a finite distributive lattice is obtained by removing a family of its irreducible intervals. The necessary direction appears in current work on maximal-sublattice complements: a maximal element of the complement of a $(0, 1)$ -sublattice is meet-irreducible and a minimal element is join-irreducible.

One difference, tested rather than assumed. The literature's primeness is *relativised* — prime within the ideal of v and the filter of u — where §27's is absolute. On Λ_8 the two agree exactly: **0 disagreements each across all 116,138 intervals**. So the paper's form is a correct specialisation and the relativised form is the general statement, which is the one to cite.

And it supplies a named property this paper had computed without naming. A lattice is *dismantlable* when it can be reduced to nothing by removing doubly irreducible elements one at a time. **Λ_8 has none, so Λ_8 is not**

dismantlable — which is §28's pushback floor of 16 and §27's step of 4 stated in the standard vocabulary.

What remains unlocated is the quantity. The literature characterises which intervals dismantle; it was not found treating **step(X) = the minimum size of a dismantling interval** as an invariant, nor the $(n_{\max} - 1)(e_{\max} - 1)$ form measured at five cap settings. That is the same narrow residue as O7 — the class is named, the graded measure is not. §27.1 applies: the absence is not evidence.

O9 — CLOSED, in both halves, and neither answer is the one the item was written to get.

The item's history. It began as *a measurement the composition order predicts and something else does not*, standing without a candidate. A candidate was proposed — no cascade raises occupancy — and **disqualified**: that claim is a consequence of §7.1's conflation of *placed* with *present*, so against real cascades the matching condition $k_b = g_a$ is false and the test cannot fail. Disqualifying it produced §11. The item was then reframed as *does any temporal structure survive occupied destinations?*

First half, answered by computation and it is negative on the whole object. In the extended index the composition graph is **one strongly connected component** over all 33 objects; every coordinate rises and falls; no monotone weight exists and none can. **What survives is the maximal non-accretive sub-index** — Part IX — which is closed, three times the size of Λ_9 , and restores the eras exactly. So the temporal structure survives *as a restriction*, and the restriction is the answer.

Second half, answered by the precedent, and it dissolves the question. Part IX identifies the construction: an arrow is a **ranking function** and the maximal sub-index is a **supporting invariant**. A ranking function is a certificate an analyst supplies from a class the analyst chooses — every terminating relation has one, and the class is what determines tractability.

A certificate has nothing to predict. Asking what the composition order forecasts that something else does not is asking what a proof technique forecasts. The measurement O9 sought does not exist, and not for want of searching.

Which is the third wall. §16.7.3 says the index cannot tell you whether it describes anything. §11 says the arrow was carried by a modelling assumption. **This says that even granting both, the object is a certificate and certificates do not predict.** The first two deferred the question; this one closes it.

Unclosable

O10 — the destination coordinate. For the temporal mechanisms to name a destination, a coordinate must exist that the index does not carry. §30.1.4 proves $\mathcal{R}(X) \subseteq \prod_i A_i(X)$, so the operator cannot produce a value the index lacks, let alone an axis; §18.1.1's Theorem 11.2 proves no ordinal structure over these coordinates carries what the coordinates do not, whatever formula the join takes, and that the only escape is a further measurement — **which is by definition a new coordinate**. P20 classifies the claim as posed in the documentary language, which has no closure mechanism, so it is **not false, not true, unclosable**.

And this appendix has now demonstrated the boundary rather than asserted it. Every other item was closed, bounded, located, or reduced to a stated act — by computation, by search, by a theorem, or by a patch. O10 alone admits none, and the reason is not difficulty. **Nine items were reachable because they ask what the index contains; O10 asks what it is written in.** Register 332 — E names cells and never coordinates — is what separates them, and this appendix is its worked example.

O10 is not an open question this paper has. It is the shape of the question the index cannot pose, and it is the only item here with no test, because a test would be the thing it says does not exist.

Appendix D — Every result in universal form

Each row states a result without reference to Λ , together with the hypothesis it needs. Throughout: X is an index — a set of cells cut from a product of finite chains by constraints $x_i \leq \varphi_{ij}(x_j)$ with φ non-decreasing; $\mathcal{R}$ is the closure operator; cells may carry *src* and *tgt* in an object set O , a weight $w : O \rightarrow \mathbb{N}$ and a transfer $t : X \rightarrow \mathbb{N}$.

#	result	universal expression	hypothesis
1	closure defect	$E(X) = \mathcal{R}(X) - X $	$X \subseteq \prod \text{chains}$
2	closure criterion	$X \text{ closed} \Leftrightarrow \mathcal{R}(X) = X$	—
3	transfer bound	$t(a) \leq w(\text{src } a)$	cells are transitions
4	destination law	$w(\text{tgt } a) = w_0(a) + t(a)$	as above

#	result	universal expression	hypothesis
5	non-accretivity	$w_0(a) \leq w(\text{src } a) - t(a) \Leftrightarrow w(\text{tgt } a) \leq w(\text{src } a)$	3, 4
6	the clock	w non-increasing along composition $\Leftrightarrow 5$	3, 4
7	cycles	every cycle lies in a level set of w ; every step conservative	6
8	the arrow	$X_w = \{a \in X : w(\text{tgt } a) \leq w(\text{src } a)\}$, the <i>maximum</i> sub-index carrying 6	3, 4
9	admissibility	X_w closed $\Leftrightarrow w$ is a projection ; then 8 is a bound of one coordinate by the identity of another, which 2 admits	$X \subseteq \prod$ chains
10	co-arrow and core	$X_w^{\text{op}} = \{w(\text{tgt}) \geq w(\text{src})\}$; $X_w \cup X_w^{\text{op}} = X$; $X_w \cap X_w^{\text{op}} = \{w(\text{tgt}) = w(\text{src})\}$, all closed	9
11	the tick	$w(\text{src } a) - w(\text{tgt } a) = [w(\text{src } a) - t(a)] + [t(a) - w(\text{tgt } a)]$	3, 5
12	extraction	$X \mapsto X_w$ is contractive, monotone, idempotent and preserves intersections — a meet-morphism	pointwise predicate
13	family algebra	$\{X_w\}$ is a meet-semilattice; joins exist in the lattice of sublattices and are not arrows	9, 12
14	total fragmentation	under all coordinates' arrows, every cycle is a self-loop	8, one arrow per coordinate
15	decay	$ P(A_u \cap A_v) - P(A_u)P(A_v) \leq \sqrt{I(u,v) \ln 2 / 2}$, and $I(u,v)$ non-increasing in tree distance	constraint graph is a tree
16	Markov property	coordinates form a Markov random field on the constraint graph	tree
17	balance	$\sum_v A(v) \cdot B(v) = X $ at every two-sided cut	tree
18	cut law	a component shrinks iff it holds a parent of the cut coordinate, grows iff a child	tree
19	step law	$X \setminus [a,b]$ is a sublattice $\Leftrightarrow a$ is join-prime and b is meet-prime	X distributive
20	step	$\text{step}(X) = \min\{ [a,b] : a \text{ join-prime, } b \text{ meet-prime, } a \leq b, 1 < [a,b] < X \}$	19
21	pushback	$J(x) = \#\{(u,v) : uvv = x, u,v \neq x\}$; $M(x)$ dual; both local to $[\perp, x]$ and $[x, \top]$	X distributive
22	removability	a single cell is removable $\Leftrightarrow$ it is doubly irreducible	19
23	meet-closure	a set defined by pointwise minima of concave functions is meet-closed	concavity
24	prediction budget	forward statement has budget $E(X)$; interpolation does not	1

Four hypotheses do all the work, and each has a falsifier that has been fired. $X \subseteq \prod$ chains with monotone φ — replacing φ by a decreasing or non-monotone function breaks 678,750 and 67,500 joins (§37). *Cells are transitions with a transfer* — without 3 and 4 there is no clock (§11, 31.6% of steps rise). *w is a projection* — sums, max and min all fail (§37). **The constraint graph is a tree** — rows 15 to 18 all fail at Λ_0' , the one stage carrying a cycle (§29, §30, §41.1).

What has no universal expression, and is not given one. The step's closed form $(n_{\text{max}}-1)(e_{\text{max}}-1)$, a pattern at five settings. The limiting densities 1, 3/5, 48/185, whose general term would come from the pattern the values make. The strict ordering of arrow overlaps, which is measured at four cap settings while only its decay is proved. Which coordinate gives the strictest arrow, cap-dependent. §20.1.2 forbids taking the next term from the terms, and these are the four places that rule bites.

Appendix E — References

- Brylawski, T. “The lattice of integer partitions.” *Discrete Mathematics* **6**(3), 201–219 (1973).
- Baker, K. A., and Pixley, A. F. “Polynomial interpolation and the Chinese remainder theorem for algebraic systems.” *Mathematische Zeitschrift* **143**, 165–174 (1975).
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- Knaster, B., and Tarski, A. Fixed-point theorem for monotone functions on a complete lattice. (Tarski, *Pacific Journal of Mathematics* **5**, 285–309, 1955.)
- Lach, M. *The Method — The Lach Cylinder: an index of transitions — closure, self-reference, refusal, and the price of certainty*. Version 1.3 (2026).

All computations in this paper are reproducible from the definitions printed in The Method 1.3 and the methods of Appendix A. A reader who rebuilds Λ and does not reproduce the counts of §2 has found an error and should say so.